Hypergraph Turán densities can be transcendental

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Abstract

We construct an integer $k \geq 3$ and a finite family $\mathcal{F}$ of finite simple k-uniform hypergraphs whose classical Turán density is transcendental. The construction is based on a finite system of literal marker profiles. A bounded positive-certificate argument converts the resulting hereditary profile class into a genuinely finite obstruction family. Its weighted extremal problem contains a partial bipartite orientation. A sharp feedback-arc repair bounds its weighted score by that of a complete Ferrers matrix, whose exact payload optimum is $1/(2e^2)$. The remaining optimization is the value of a rational polynomial over the rational symmetric simplex $\mathcal{K}$, after a separately stated numerical relaxation on $\mathcal{D}$. Its value function is a strictly increasing semialgebraic function. An algebraic-curve specialization then shows that its value at $1/(2e^2)$ is transcendental. A finite core-pair completion transfers the weighted identity exactly to ordinary non-induced Turán containment.

1 Introduction

An r-uniform hypergraph, or r-graph, is a finite set of r-subsets of a finite vertex set. For a family $\mathcal{H}$ of r-graphs, let $\mathrm{ex}(n, \mathcal{H})$ be the largest number of edges in an n-vertex $\mathcal{H}$-free r-graph, where containment is ordinary and non-induced. Its Turán density is

$$\pi(\mathcal{H}) = \lim_{n\to\infty} \frac{\mathrm{ex}(n, \mathcal{H})}{\binom{n}{r}}.$$

The limit exists by the averaging argument of Katona, Nemetz and Simonovits [7].

The algebraic complexity of hypergraph Turán densities is poorly understood. Baber and Talbot, and independently Pikhurko, produced irrational densities of finite forbidden families; Pikhurko's construction works in every uniformity at least three [2, 10]. Pikhurko also recorded a question of Jacob Fox asking whether a finite forbidden family can have a transcendental Turán density [10, Question 27]. Grosu subsequently developed the algebraic and topological structure of the sets of Turán densities [5], and Liu and Pikhurko constructed finite-family densities of arbitrarily large algebraic degree [8]. Related transcendental constants occur in other extremal settings: Hou, Li, Liu, Mubayi and Zhang exhibited extremal constructions with a transcendental *shadow* density [6], while Yuster obtained the value $2/e^2$ for the inducibility of P_4 in ordered monotone balanced bipartite graphs [11]. These are not classical non-induced Turán densities of finite hypergraph families.

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Recent work of Liu and Pikhurko shows that densities of possibly infinite forbidden families contain non-degenerate intervals [9]. To the best of our knowledge at the time of submission, none of these results produces a transcendental classical Turán density from one fixed finite forbidden family. We prove that such a family exists.

Theorem 1.1. *There are an integer $k \geq 3$ and a finite family $\mathcal{F}$ of finite simple k-graphs such that*

$$\pi(\mathcal{F})$$

is transcendental over $\mathbb{Q}$.

We take the explicit uniformity $k = 1000$; a rational certificate for all quantitative inequalities is given in Section 11, with the complete cross-multiplied integer certificate and its machine-readable implementation in Appendix A. Let $q = k^3$, and let $\mathscr{C}$ be the class encoded by the finite marker profiles in Section 3. We prove that

$$\mathscr{C} = \mathrm{Forb}(\mathcal{A})$$

for an actual finite family $\mathcal{A}$. A second finite family $\mathcal{F}$, consisting of bounded core-pair completions of the members of $\mathcal{A}$, satisfies

$$\pi(\mathcal{F}) = k!\Lambda(\mathscr{C}), \tag{1.1}$$

where $\Lambda(\mathscr{C})$ is the supremum of the unscaled Lagrangians of members of $\mathscr{C}$.

The weighted problem in $\mathscr{C}$ contains a partial 0–1 matrix with no checkerboard. The weighted repair lemma shows that, after changing cells of total weight no larger than the missing-cell weight, completion to a Ferrers matrix produces a numerical score no smaller than the original one. The state gain pays, with a fixed positive margin, for every payload destroyed by a missing or changed cell. For a Ferrers boundary g, the normalized payload is

$$\mathcal{T}(g) = \int_{0<x<y<1} g(y)\big(g(x) - g(y)\big)\,dx\,dy.$$

Its exact maximum is

$$\tau_0 = \frac{1}{2e^2}. \tag{1.2}$$

After exact symmetrization of the marker weights, the remaining optimization has the form

$$\Lambda(\mathscr{C}) = \Phi(\tau_0), \tag{1.3}$$

where $\Phi(t)$ is the maximum of a polynomial in t and finitely many weight variables, with rational coefficients, over a rational compact simplex. The function Φ is semialgebraic over $\mathbb{Q}$ and strictly increasing. Since τ_0 is transcendental by the Lindemann–Weierstrass theorem [3, Chapter 1, §3], an algebraic-curve specialization argument shows that $\Phi(\tau_0)$ is transcendental. Equations (1.1)–(1.3) prove the theorem.

The proof is deliberately finite at both interfaces. The local profile rules do not stand for an infinite forbidden family: the search-tree argument gives a numerical obstruction order. Likewise, the passage from the weighted problem to the classical density uses a bounded completion family and not a limiting family-extension theorem. The four interfaces specific to the proof are a finite certificate-positive forcing mechanism, an exact core-pair-completion transfer, a sharp weighted score comparison with a Ferrers matrix, and a semialgebraic specialization at the transcendental Ferrers optimum.

A key point is that the transcendental number is never inserted into the definition of the forbidden family. All profiles, obstruction orders and completion bounds are integer data. The constant $1/(2e^2)$ appears only after solving an internal Ferrers variational problem, and the remaining optimization has rational coefficients. Thus the finite-family assertion is separated from the transcendental specialization rather than being encoded by a real parameter.

2 Preliminaries and notation

For a finite set X and $j \geq 0$, write $\binom{X}{j}$ for its j-subsets. For nonnegative variables $(z_i)_{i\in I}$, let

$$e_j(z) = \sum_{S\in\binom{I}{j}} \prod_{i\in S} z_i$$

be the jth elementary symmetric polynomial. If G is a finite k-graph, put

$$p_G(x) = \sum_{e\in E(G)} \prod_{v\in e} x_v, \qquad \lambda(G) = \max\left\{p_G(x) : x_v \geq 0,\ \sum_v x_v = 1\right\}.$$

This is the unscaled Lagrangian; the frequently used scaled convention is $k!\lambda(G)$. A blow-up of G replaces each vertex by a cluster and each edge by the complete transversal k-partite k-graph across the corresponding clusters. If the cluster proportions tend to x, then its edge density tends to $k!p_G(x)$.

We use $L = 4$ cyclic state profiles and $J = 100$ tag profiles. The uniformity k is taken sufficiently large during the estimates and is then frozen by the rational prescription in Section 11; throughout the construction

$$q = k^3.$$

All indices i attached to A_i or B_i are read modulo L.

Object-level terminology. An *actual graph* or *template* is always a finite simple k-graph, together with a legal marker map when one is under discussion. A *weighting* is a point of the vertex simplex of such a graph. A *partial state matrix* is extracted only from the positive-weight row and column objects of an actual weighted template. By contrast, a *relaxed tuple* is a numerical point $(y, a, b, E_0, E_1) \in \mathcal{D}$; it need not be realized by any graph. Finally, a point of $\mathcal{K}$ is a symmetric numerical weight point. These terms are kept distinct below; the complete object and quantifier ledger appears before Proposition 12.2.

The argument is organized as follows. Section 3 defines the literal profile class and proves its finite obstruction basis; Section 4 gives the exact finite core-completion transfer. The matrix part of the weighted problem is solved from Section 5 through Section 7. The marker weights are localized and symmetrized in Section 9 and Section 10. The algebraic envelope and transcendence are established in Section 12 and Section 13, while Section 14 records the finiteness and extremal audits.

3 Literal marker profiles and a finite forbidden family

In this section we turn the finite collection of profile rules used in the variational construction into a genuinely finite family of ordinary simple k-graphs. There are two separate points that must not be conflated. First, the marker labels are optional: after taking a subgraph, the labels whose vertices have disappeared are simply forgotten. Second, all local rules are *certificate-positive*: after the earlier uniqueness rules are taken into account, every violation—including the exact-pattern rules (R4)–(R6)—is certified by the presence of a bounded number of edges. These two features imply, by a bounded search-tree argument, that the represented class has a finite obstruction basis for ordinary, non-induced containment.

Throughout this section, $k \geq 6$ is fixed. We also fix positive integers L and J; in the application we take

$$L = 4 \qquad \text{and} \qquad J = 100.$$

Let

$$q \geq 2k - 6 + 2L + 2J$$

be fixed, and let $\mathcal{M}$ be a set of q formal marker labels. Inside $\mathcal{M}$ choose labels and label sets

$$h, \quad H, \quad D, \quad A_i, B_i \ (i \in \mathbb{Z}/L\mathbb{Z}), \quad p_j, p'_j \ (1 \leq j \leq J)$$

such that

$$|H| = k - 2, \qquad h \in H, \qquad |D| = k - 4,$$

and such that the following sets are pairwise disjoint:

$$\begin{aligned} &\{h\}, \quad H \setminus \{h\}, \quad D, \\ &\{A_i : i \in \mathbb{Z}/L\mathbb{Z}\}, \quad \{B_i : i \in \mathbb{Z}/L\mathbb{Z}\}, \\ &\{p_j : 1 \leq j \leq J\}, \quad \{p'_j : 1 \leq j \leq J\}. \end{aligned}$$

The displayed lower bound on q is exactly the number of reserved labels. In the application, $q = k^3$, and k is fixed sufficiently large that this inequality and the analytic inequalities used later all hold. Any remaining elements of $\mathcal{M}$ are auxiliary marker labels. All parameters in this definition are integers. In particular, neither the class below nor any of its finite obstruction families uses a real parameter.

3.1 Partial marker maps and permitted edge profiles

Let G be an ordinary simple k-graph. A *partial marker map* on G is an injection

$$\mu : \operatorname{dom}\mu \longrightarrow V(G), \qquad \operatorname{dom}\mu \subseteq \mathcal{M}.$$

The vertices in $\operatorname{im}\mu$ are called *marked vertices*; all other vertices are called *objects*. We write

$$O_\mu := V(G) \setminus \operatorname{im}\mu.$$

If $X \subseteq \operatorname{dom}\mu$, then $\mu(X)$ denotes the set of marked vertices carrying the labels in X. No label is required to be present.

An edge $e \in \binom{V(G)}{k}$ is *profile-admissible with respect to* μ if it has one of the following forms. A displayed profile is available only when every marker label displayed in it lies in $\operatorname{dom}\mu$.

(P0) e consists of k marked vertices. Thus every all-marker k-set is permitted.

(PN) There are $x \in O_\mu$ and $S \in \binom{\operatorname{dom}\mu \setminus \{h\}}{k-1}$ such that

$$e = \{x\} \cup \mu(S).$$

These are the neutral one-object profiles.

(PR) There are $x \in O_\mu$ and $j \in [J]$ such that

$$e = \{x\} \cup \mu(H \cup \{p_j\}).$$

(PC) There are $x \in O_\mu$ and $j \in [J]$ such that

$$e = \{x\} \cup \mu(H \cup \{p'_j\}).$$

(PSε) There are distinct $x, y \in O_\mu$ and $i \in \mathbb{Z}/L\mathbb{Z}$ such that, for $\varepsilon \in \{0,1\}$,

$$e = \{x, y\} \cup \mu\big(D \cup \{A_i, B_{i+\varepsilon}\}\big).$$

(PP) There is $X \in \binom{O_\mu}{4}$ such that

$$e = X \cup \mu(D).$$

The cardinalities in the definition are exactly k: for example, $|D| + 2 + 2 = k$ in (PSε), and $|D| + 4 = k$ in (PP). Notice also that a neutral profile never contains the label h, whereas each row or column tag profile contains h through H. Thus (PN) does not overlap (PR) or (PC).

The following predicates are declared by actual edges of G. For $x \in O_\mu$, write $R_\mu(x)$ if

$$\{x\} \cup \mu(H \cup \{p_j\}) \in E(G) \quad \text{for some } j \in [J],$$

and write $C_\mu(x)$ if

$$\{x\} \cup \mu(H \cup \{p'_j\}) \in E(G) \quad \text{for some } j \in [J].$$

For distinct $x, y \in O_\mu$ and $\varepsilon \in \{0,1\}$, write $S^\varepsilon_\mu(x, y)$ if

$$\{x, y\} \cup \mu\big(D \cup \{A_i, B_{i+\varepsilon}\}\big) \in E(G) \quad \text{for some } i \in \mathbb{Z}/L\mathbb{Z}.$$

Finally, for $X \in \binom{O_\mu}{4}$, write $P_\mu(X)$ if

$$X \cup \mu(D) \in E(G).$$

As above, a predicate is false if one of the marker labels occurring in its definition is absent. We call an object *tagged* if it satisfies R_μ or C_μ, and call a pair *stated* if it satisfies S^0_μ or S^1_μ.

The word "declared" is literal here: one physical profile edge is enough to declare a tag or a state. The definition never uses the absence of an edge. In particular, the profiles above are permissions, not requirements; no permitted edge is forced to be present.

3.2 The positive local rules

A partial marker map μ is called *legal for G* if every edge of G is profile-admissible with respect to μ and the following six rules hold.

(R1) No object has both tags:

$$\neg\big(R_\mu(x) \wedge C_\mu(x)\big).$$

(R2) No object-pair has both states:

$$\neg\big(S^0_\mu(x, y) \wedge S^1_\mu(x, y)\big).$$

(R3) If a stated pair has two tagged endpoints, then one endpoint is a row object and the other is a column object. Equivalently, the patterns

$$R_\mu(x) \wedge R_\mu(y) \wedge S^\varepsilon_\mu(x, y) \quad \text{and} \quad C_\mu(x) \wedge C_\mu(y) \wedge S^\varepsilon_\mu(x, y)$$

are forbidden for $\varepsilon \in \{0,1\}$.

(R4) There is no checkerboard on two row objects and two column objects. More precisely, let r_1, r_2, c_1, c_2 be pairwise distinct, with $R_\mu(r_a)$ and $C_\mu(c_b)$ for $a, b \in \{1, 2\}$. If all four cross-pairs are stated, neither of the following two state matrices is allowed:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Here the (a, b) entry is ε when $S^\varepsilon_\mu(r_a, c_b)$ holds. Rule (R2) makes this entry unique.

(R5) If $P_\mu(X)$ holds and all four elements of X are tagged, then precisely two elements of X have the row tag and precisely two have the column tag.

(R6) Let r_1, r_2, c_1, c_2 be pairwise distinct row, row, column and column objects, respectively. Suppose that

$$P_\mu(\{r_1, r_2, c_1, c_2\})$$

holds and that all four cross-pairs are stated. Then exactly one of those four cross-pairs has state 1.

Let $\mathscr{C} = \mathscr{C}(k, q, L, J)$ be the class of all finite simple k-graphs G for which there exists at least one partial marker map legal for G. The quantifier over marker maps is existential. Different members of $\mathscr{C}$ need not use the same subset of marker labels.

Lemma 3.1 (Monotonicity under ordinary containment). *If $G \in \mathscr{C}$ and G' is an ordinary, not necessarily induced, subgraph of G, then $G' \in \mathscr{C}$.*

Proof. After relabelling vertices, assume that $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$. Let μ be legal for G, and restrict it to

$$\operatorname{dom} \mu' := \{m \in \operatorname{dom} \mu : \mu(m) \in V(G')\}.$$

Thus μ' is obtained by forgetting precisely those labels whose marked vertices were deleted. Every surviving vertex has the same marked/object status and, if marked, the same label under μ' as under μ. Consequently every edge of G' has exactly the same profile under μ' as it had under μ, and is therefore profile-admissible. Moreover, every tag, state or payload predicate witnessed in G' is witnessed by the same physical edge in G. Since every forbidden local pattern is certified by a positive collection of such predicates, deleting edges cannot create a violation. Thus μ' is legal for G'. □

The use of partial, rather than total, marker maps is essential in the preceding proof. A definition requiring all q marker vertices would not be closed under deleting one of them.

3.3 A bounded certificate for non-representability

For a fixed partial marker map μ, failure of legality has a small positive certificate. If an edge is not profile-admissible, that edge alone is a certificate. Otherwise inspect (R1)–(R6) in order and choose the first failed rule. For that rule choose one physical edge witnessing each tag, state and payload predicate that occurs in the violation. In particular, if (R5) is the first failed rule, then (R1) holds, so each of the four tagged objects has a unique tag and four tag witnesses suffice. The numbers of witnesses needed for an inadmissible edge and for (R1)–(R6), respectively, are at most

$$1,\ 2,\ 2,\ 3,\ 8,\ 5,\ 9.$$

The largest case is (R6), which requires one payload edge, four tag edges and four state edges. Consequently every failure has a certificate consisting of at most nine edges, and hence involving at most

$$w := 9k$$

vertices.

We record the persistence property that makes the argument work. If μ and ν are partial marker maps, write $\mu \preccurlyeq \nu$ when $\operatorname{dom}\mu \subseteq \operatorname{dom}\nu$ and ν agrees with μ on $\operatorname{dom}\mu$.

Lemma 3.2 (Persistence of a positive certificate)**.** *Let W be the union of the edges in the certificate selected above for the first failed condition under a partial marker map μ on G. Let H be any subgraph of G containing every certificate edge, and let ν be a partial marker map on H with $\mu \preccurlyeq \nu$. If*

$$\operatorname{im}\big(\nu|_{\operatorname{dom}\nu\setminus\operatorname{dom}\mu}\big) \cap V(W) = \varnothing,$$

then ν is not legal for H. Consequently, every legal extension of μ on a subgraph containing the certificate edges assigns an unused marker label to a vertex of $V(W) \setminus \operatorname{im}\mu$.

Proof. Every vertex of a certificate edge has the same marked/object status under ν as under μ, and every old marked vertex has the same label. Thus each certificate edge has the same profile, and every tag, state or payload predicate witnessed by a certificate edge under μ remains true under ν. Other edges of H, together with the new labels, may create further predicates, so for the rules involving an exact pattern we check that this cannot restore legality.

If the certificate is an inadmissible edge, its unchanged profile is still inadmissible. If the first failed rule is (R1), (R2) or (R3), the same positive forbidden collection of predicates persists, so that rule still fails. Suppose next that the first failed rule is (R4). The four old tags and four old states persist. If one of the four objects has acquired the opposite tag, then (R1) fails; if one of the four cross-pairs has acquired the opposite state, then (R2) fails. Otherwise the original checkerboard, with the same row/column roles and the same four unique states, persists and (R4) fails.

If the first failed rule is (R5), then (R1) held under μ, so each of the four certificate objects had a unique old tag. An acquired opposite tag violates (R1); if none is acquired, the old non-2–2 tag split persists and (R5) still fails. Finally suppose that the first failed rule is (R6). An acquired opposite tag or opposite state violates (R1) or (R2), respectively. If neither occurs, the old row–row–column–column roles and the old number of state-1 cells persist, so (R6) still fails. In every case ν is non-legal. The final assertion is the contrapositive. $\square$

Proposition 3.3 (Finite basis)**.** *There is an integer $N = N(k, q, L, J)$ and a finite family $\mathcal{A}$ of finite simple k-graphs, each on at most N vertices, such that*

$$\mathscr{C} = \operatorname{Forb}(\mathcal{A}),$$

where Forb *refers to ordinary non-induced containment. The family $\mathcal{A}$ depends only on the integer data k, q, L, J and the finite label sets fixed above.*

Proof. Let $G \notin \mathscr{C}$. We construct a rooted search tree of partial marker maps. The root is the empty map. At a node μ, choose a certificate W_μ of non-legality using at most nine edges. For every

$$m \in \mathcal{M} \setminus \operatorname{dom}\mu, \qquad v \in V(W_\mu) \setminus \operatorname{im}\mu,$$

add the child $\mu \cup \{m \mapsto v\}$. Because m is unused and $v \notin \operatorname{im}\mu$, every child is again a partial injection. These are all possible one-label extensions that can affect the certificate. Since G has no legal partial marker map, no node is legal. Every branch has length at most q, and each node has at most

$$B := qw = 9kq$$

children. Thus the tree has at most

$$T := 1 + B + B^2 + \cdots + B^q$$

nodes.

Let K be the subgraph of G whose edges are all certificate edges selected at all nodes of this tree, and whose vertex set is the union of those edges. Then

$$|V(K)| \leq wT.$$

Every vertex newly inserted into the image of a child map lies in $V(W_\mu)$ for its parent node μ. Since every W_μ is one of the selected certificates, induction from the empty root shows that

$$\operatorname{im}\mu \subseteq V(K) \qquad \text{for every node map } \mu \text{ in the search tree.} \tag{3.1}$$

This inclusion is what permits the persistence lemma to be applied with $H = K$ at every stage.

We claim that $K \notin \mathscr{C}$. Suppose, to the contrary, that η is a legal partial marker map for K. At the root, η extends the empty map. More generally, suppose that η extends a node μ on the chosen root-to-node path. By (3.1), both maps are maps on K, and the certificate W_μ is a subgraph of K. Apply Lemma 3.2 with the original ambient graph G, with $H = K$, with old map μ, and with extending map η. Since η is legal, the contrapositive conclusion of that lemma gives an unused label $m \in \operatorname{dom}\eta \setminus \operatorname{dom}\mu$ whose image $v = \eta(m)$ lies in $V(W_\mu) \setminus \operatorname{im}\mu$. Hence the child $\mu \cup \{m \mapsto v\}$ occurs in the search tree, and η extends it.

Repeating this step increases the domain by one each time and therefore, after exactly $|\operatorname{dom}\eta|$ steps, reaches the node map $\mu = \eta$. At this terminal node, apply Lemma 3.2 once more with the old map $\mu := \eta$ regarded as a node map on G, with $H = K$, and with the extending map $\nu := \eta$ regarded as a map on K. The new-label image is empty, so the disjointness hypothesis is vacuous; because K contains every edge of W_η, the lemma says that η is not legal for K. This contradicts the choice of η and proves the claim.

Set

$$N := w(1 + B + \cdots + B^q).$$

Membership in $\mathscr{C}$ is invariant under graph isomorphism: a legal partial marker map transports along any graph isomorphism. Let $\mathcal{A}$ contain one representative of every isomorphism class of simple k-graphs A with $|V(A)| \leq N$ and $A \notin \mathscr{C}$. This is a genuinely finite family: there are only finitely many simple k-graphs on at most N vertices, and membership in $\mathscr{C}$ is decided by checking the finitely many partial injections from subsets of $\mathcal{M}$. This is only a finite decidability argument; no efficiency claim is made for this exhaustive enumeration.

If $G \in \mathscr{C}$, then Lemma 3.1 shows that G contains no member of $\mathcal{A}$: from an ordinary copy one may retain exactly the copied edges and discard all additional edges on its image. Conversely, if $G \notin \mathscr{C}$, the preceding search-tree construction produces a subgraph $K \subseteq G$ with $|V(K)| \leq N$ and $K \notin \mathscr{C}$; hence G contains a member of $\mathcal{A}$. This proves the asserted equality. $\square$

It is worth stressing what Proposition 3.3 does and does not say. It does not replace an infinite family by a finite algorithm describing that family. It gives a numerical vertex bound N and takes a subset of the finite set of all k-graphs on at most N vertices. Thus $\mathcal{A}$ is finite in the literal sense required for the ordinary Turán problem.

4 The bounded core-pair completion transfer

The family $\mathcal{A}$ obtained above is the correct family for the Lagrangian optimization, but blow-ups of an $\mathcal{A}$-free graph need not themselves be $\mathcal{A}$-free. We now replace $\mathcal{A}$ by another finite family for which blow-ups give the lower bound, while a minimal-support Lagrangian argument gives the upper bound. This step is entirely finite and uses no removal lemma or compactness approximation.

Recall the unscaled Lagrangian $\lambda(G)$ from Section 2. For a monotone class $\mathscr{D}$ of simple k-graphs, write

$$\Lambda(\mathscr{D}) := \sup\{\lambda(G) : G \in \mathscr{D}\}.$$

Let A be a simple k-graph and let $f = |V(A)|$. Define

$$b(A) := f + (k-2)\binom{f}{2}.$$

An *A-core completion* is a simple k-graph Q on at most $b(A)$ vertices for which there exists an injection

$$\iota : V(A) \longrightarrow V(Q)$$

with the following properties:

1. $\iota(e) \in E(Q)$ for every $e \in E(A)$;
2. for every two distinct $a, b \in V(A)$, there is an edge of Q containing both $\iota(a)$ and $\iota(b)$.

The image $\iota(V(A))$ is the distinguished core. The distinction is used only to define the family; the forbidden object is the underlying ordinary unlabelled k-graph Q.

Let $\mathrm{Comp}(A)$ contain one representative of every isomorphism class of underlying A-core completions, and define

$$\mathcal{F} := \bigcup_{A \in \mathcal{A}} \mathrm{Comp}(A).$$

This is a finite family of ordinary simple k-graphs. Indeed, $\mathcal{A}$ is finite and every member of $\mathrm{Comp}(A)$ has at most $b(A)$ vertices. The family is also nonempty for each A: one may cover each core pair by a k-edge using $k-2$ new private vertices. No marker labels, colors, orientations or weights are retained in $\mathcal{F}$.

We use the standard blow-up notation. If G is a simple k-graph and $\boldsymbol{n} = (n_v)_{v \in V(G)}$ is a vector of nonnegative integers, then $G(\boldsymbol{n})$ is obtained by replacing v with a cluster V_v of size n_v and, for every $e \in E(G)$, inserting every k-set having one vertex in each cluster V_v, $v \in e$. Thus every edge of a blow-up meets k distinct clusters.

Lemma 4.1 (Blow-up lower-bound lemma)**.** *If G is $\mathcal{A}$-free, then every blow-up of G is $\mathcal{F}$-free.*

Proof. Suppose that a copy $\varphi : V(Q) \to V(G(\boldsymbol{n}))$ of some $Q \in \mathrm{Comp}(A)$ occurs in a blow-up $G(\boldsymbol{n})$, where $A \in \mathcal{A}$. Fix one core injection $\iota : V(A) \to V(Q)$ witnessing that Q is an A-core completion, and let $\rho : V(G(\boldsymbol{n})) \to V(G)$ be the map sending each blow-up vertex to its cluster index. Any two core vertices of Q lie together in an edge of Q. Since every blow-up edge meets distinct clusters, their images under $\rho \circ \varphi$ are distinct. Hence $\rho \circ \varphi$ is injective on the image of the core. For each $e \in E(A)$, the corresponding edge of Q projects to an edge of G. Therefore $\rho \circ \varphi \circ \iota$ is a copy of A in G, a contradiction. $\square$

The upper bound uses the following elementary support observation.

Lemma 4.2 (Pair-covered optimal support)**.** *Let H be a finite simple k-graph with $\lambda(H) > 0$. Among all optimal weight vectors for H, choose one, $\boldsymbol{x}$, whose positive support $S = \{v : x_v > 0\}$ has minimum cardinality. Then every pair of distinct vertices of S is contained in an edge of the induced subgraph $H[S]$.*

Proof. Suppose that $u, v \in S$ are not contained together in an edge of $H[S]$. Keep all other coordinates fixed and keep $x_u + x_v = s$ fixed. On the interval $0 \leq t \leq s$, replace (x_u, x_v) by $(t, s-t)$. Every edge containing a vertex outside S contributes zero, and no edge of $H[S]$ contains both u and v. Hence the Lagrangian polynomial on this interval is affine in t. Its maximum is attained at an endpoint. Replacing $\boldsymbol{x}$ by that endpoint gives another optimal vector with smaller positive support, contrary to the choice of S. □

Lemma 4.3 (Universal Lagrangian upper bound)**.** *If H is $\mathcal{F}$-free, then*

$$\lambda(H) \leq \Lambda(\mathscr{C}), \qquad \mathscr{C} = \mathrm{Forb}(\mathcal{A}).$$

Proof. The assertion is immediate when $\lambda(H) = 0$, so suppose that $\lambda(H) > 0$. Choose a minimum-support optimal vector as in Lemma 4.2, let S be its support, and put $D = H[S]$. The graph D is pair-covered.

We claim that D is $\mathcal{A}$-free. Otherwise, fix a copy of some $A \in \mathcal{A}$ in D. For every unordered pair of vertices in this core, choose an edge of D containing the pair. Let Q be the subgraph consisting of the core edges of A and all the chosen covering edges, with vertex set the union of the core and these edges. If $f = |V(A)|$, then

$$|V(Q)| \leq f + (k-2)\binom{f}{2} = b(A).$$

Thus Q is an A-core completion and belongs, up to isomorphism, to $\mathcal{F}$. Since $Q \subseteq D \subseteq H$, this contradicts the $\mathcal{F}$-freeness of H. Hence $D \in \mathscr{C}$.

All coordinates outside S have weight zero, so restricting the chosen optimal vector to D gives $\lambda(D) \geq \lambda(H)$. Conversely, every weighting of D extends by zero outside S to a weighting of H with the same value, and hence $\lambda(H) \geq \lambda(D)$. Therefore

$$\lambda(H) = \lambda(D) \leq \Lambda(\mathscr{C}).$$

□

Theorem 4.4 (Exact finite-family transfer)**.** *Let $\mathscr{C} = \mathrm{Forb}(\mathcal{A})$ be the profile class from Proposition 3.3, and let $\mathcal{F}$ be the finite family of core completions defined above. Then*

$$\pi(\mathcal{F}) = k!\,\Lambda(\mathscr{C}).$$

Here λ and Λ use the unscaled convention of Section 2; the factor $k!$ appears only when the n^{-k} normalization is converted to $\binom{n}{k}^{-1}$. In particular, if the variational analysis proves

$$\Lambda(\mathscr{C}) = \Phi(\tau_0),$$

then the ordinary classical Turán density of the one fixed finite family $\mathcal{F}$ is exactly

$$\pi(\mathcal{F}) = k!\Phi(\tau_0).$$

Proof. Set $\Lambda = \Lambda(\mathscr{C})$. For the upper bound, let H be an $\mathcal{F}$-free k-graph on n vertices. The uniform weighting gives

$$\frac{|E(H)|}{n^k} \leq \lambda(H) \leq \Lambda$$

by Lemma 4.3. Therefore

$$\frac{|E(H)|}{\binom{n}{k}} \leq \frac{n^k}{\binom{n}{k}} \Lambda = k!\Lambda + O_k(n^{-1}).$$

Taking the maximum over H and then the limit gives $\pi(\mathcal{F}) \leq k!\Lambda$.

For the reverse inequality, fix $G \in \mathscr{C}$ and a probability vector $\boldsymbol{x} = (x_v)_{v \in V(G)}$. For every sufficiently large n, choose nonnegative integers n_v with

$$\sum_v n_v = n, \qquad \frac{n_v}{n} = x_v + O_G(n^{-1}).$$

By Lemma 4.1, the blow-up $G(\boldsymbol{n})$ is $\mathcal{F}$-free. Moreover,

$$\begin{aligned}
|E(G(\boldsymbol{n}))| &= \sum_{e \in E(G)} \prod_{v \in e} n_v \\
&= n^k \left(\sum_{e \in E(G)} \prod_{v \in e} x_v + O_G(n^{-1}) \right).
\end{aligned}$$

It follows that

$$\liminf_{n \to \infty} \frac{\mathrm{ex}(n, \mathcal{F})}{\binom{n}{k}} \geq k! \sum_{e \in E(G)} \prod_{v \in e} x_v.$$

First maximize over $\boldsymbol{x}$ and then take the supremum over $G \in \mathscr{C}$. This yields $\pi(\mathcal{F}) \geq k!\Lambda$, and completes the proof. ☐

4.1 Finiteness and parameter audit

For later reference, we summarize the exact logical output of the two constructions.

1. The uniformity is the single fixed integer k.
2. Both $\mathcal{A}$ and $\mathcal{F}$ are finite collections of finite simple k-graphs. They are unlabelled when used as forbidden graphs.
3. Containment is ordinary non-induced containment throughout.
4. The families depend only on the fixed integer data k, q, L, J and the finite choice of marker labels. They do not depend on n, a precision parameter, a truncation depth, τ_0, or any real input.
5. Proposition 3.3 is exact: the bounded search tree produces an actual finite obstruction family, not an infinite family with a finite description.
6. Theorem 4.4 is also exact. The lower bound is given by blow-ups for every sufficiently large n, while the upper bound applies to every $\mathcal{F}$-free k-graph. The factor $k!$ is exactly the conversion from the Lagrangian normalization n^k to $\binom{n}{k}$.

The sole interface with the analytic part of the proof is the identity $\Lambda(\mathscr{C}) = \Phi(\tau_0)$. Nothing in the present section proves that identity; it must be established by the universal structural and variational upper bound. Conversely, once that identity is proved, the finite-family realization and the passage to the classical Turán density require no additional forcing, compactness, or removal assertion.

5 The Ferrers variational problem

Put

$$\tau_0 := \frac{1}{2e^2}.$$

The following elementary variational lemma is the source of the transcendental parameter in our construction. We give all details, in particular fixing the convention that the Ferrers boundary describes the entries of state 0.

Lemma 5.1 (Ferrers variational lemma)**.** *Let $g\colon [0,1] \to [0,1]$ be a non-increasing measurable function and set*

$$\mathcal{T}(g) := \int_{0<x<y<1} g(y)\big(g(x) - g(y)\big)\,dx\,dy.$$

Then

$$\mathcal{T}(g) \le \tau_0.$$

Equality holds, up to changes on a null set, precisely for

$$g_*(y) = \min\left\{1, \frac{1}{ey}\right\}.$$

Consequently, the supremum is attained and is equal to $1/(2e^2)$.

Proof. Write $A := \int_0^1 g$. Fubini's theorem gives

$$\begin{aligned}\mathcal{T}(g) &= \int_{0<x<y<1} g(x)g(y)\,dx\,dy - \int_{0<x<y<1} g(y)^2\,dx\,dy \\ &= \frac{A^2}{2} - \int_0^1 yg(y)^2\,dy. \end{aligned} \tag{5.1}$$

We first fix A. Among all measurable $f\colon [0,1] \to [0,1]$ with $\int f = A$, the functional $\int_0^1 yf(y)^2\,dy$ is minimized by

$$f_c(y) := \min\{1, c/y\}, \tag{5.2}$$

where $c \in [0,1]$ is uniquely determined by

$$A = c\big(1 + \log(1/c)\big). \tag{5.3}$$

Here the value at $c = 0$ is defined by continuity. Indeed, $\theta(c) := c(1 + \log(1/c))$ is continuous on $[0,1]$, satisfies $\theta(0) = 0$ and $\theta(1) = 1$, and has derivative $\theta'(c) = \log(1/c) > 0$ on $(0,1)$. Thus (5.3) has exactly one solution for each $A \in [0,1]$. At $c = 0$ one has $A = 0$ and hence $f = 0$ almost everywhere; at $c = 1$ one has $A = 1$ and hence $f = 1$ almost everywhere. For completeness, let

$c \in (0,1)$ satisfy (5.3). On $(c,1]$ we have $yf_c(y) = c$, while on $[0,c]$ the upper constraint $f \le 1$ is active. The pointwise convexity inequality

$$yf(y)^2 - yf_c(y)^2 \ge \begin{cases} 2c\big(f(y) - f_c(y)\big), & y > c, \\ 2y\big(f(y) - 1\big), & y \le c \end{cases}$$

and the facts $f(y) - 1 \le 0$ and $2y \le 2c$ on $[0,c]$ imply

$$\int_0^1 y\big(f(y)^2 - f_c(y)^2\big)\,dy \ge 2c\int_0^1 (f - f_c)\,dy = 0.$$

Strict convexity away from the null point $y = 0$ gives uniqueness up to changes on a null set. Together with the two endpoint cases, this proves the fixed-mass assertion for every $A \in [0,1]$. Notice also that f_c is non-increasing, so the same minimizer is valid after the monotonicity constraint is imposed.

Put $u := \log(1/c)$. Direct integration yields

$$\int_0^1 yf_c(y)^2\,dy = \frac{c^2}{2} + c^2u, \qquad A = c(1+u).$$

It follows from (5.1) that

$$\mathcal{T}(g) \le \frac{c^2u^2}{2} = \frac{1}{2}e^{-2u}u^2 \le \frac{1}{2e^2}.$$

The last function has its unique positive maximum at $u = 1$, that is, $c = e^{-1}$. The equality statement follows from the uniqueness in the two optimization steps. □

We record the finite weighted form that will be used below. Let R and C be disjoint finite sets carrying non-negative weights $(x_r)_{r\in R}$ and $(y_c)_{c\in C}$, and put

$$a := \sum_{r\in R} x_r, \qquad b := \sum_{c\in C} y_c.$$

A 0–1 matrix $\sigma\colon R \times C \to \{0,1\}$ is called *Ferrers* if the zero-neighbourhoods

$$N_0(r) := \{c \in C : \sigma(r,c) = 0\}$$

are linearly ordered by inclusion. Equivalently, after suitable orders of R and C, the zeroes in every row form an initial interval and the lengths of these intervals are non-increasing.

Corollary 5.2 (Weighted Ferrers bound)**.** *If σ is Ferrers, then*

$$\sum_{\substack{\{r,r'\}\in\binom{R}{2},\ \{c,c'\}\in\binom{C}{2} \\ |\{(u,v)\in\{r,r'\}\times\{c,c'\}:\sigma(u,v)=1\}|=1}} x_r x_{r'} y_c y_{c'} \le \tau_0 a^2 b^2. \tag{5.4}$$

Moreover, the constant is best possible, even when all row weights are equal and all column weights are equal.

Proof. The assertion is trivial if $ab = 0$, so assume $a, b > 0$. Replace every row r by an interval of length x_r/a and every column c by an interval of length y_c/b. Order the rows as $r_1, \dots, r_{|R|}$ so that $N_0(r_1) \supseteq \cdots \supseteq N_0(r_{|R|})$. For a column c, let $d(c)$ be the largest index j for which $c \in N_0(r_j)$, with $d(c) = 0$ if the column lies in no zero-neighbourhood. Ordering columns by non-increasing $d(c)$

makes every $N_0(r_j) = \{c : d(c) \geq j\}$ an initial interval. Use these orders for the row and column intervals. This produces a non-increasing step function $g\colon [0,1] \to [0,1]$.

For two rows $x < y$, the columns split into three intervals on which the two entries are, respectively,

$$00, \qquad 01, \qquad 11,$$

of lengths $g(y)$, $g(x) - g(y)$, and $1 - g(x)$. Thus a pair of columns gives exactly one entry of state 1 precisely when one column is chosen from the first interval and one from the second. Consequently, with the sum taken over exactly the quadruples displayed in (5.4), one has the normalization identity

$$\frac{1}{a^2 b^2} \sum_{\substack{\{r,r'\} \in \binom{R}{2},\ \{c,c'\} \in \binom{C}{2} \\ |\{(u,v) \in \{r,r'\} \times \{c,c'\} : \sigma(u,v)=1\}| = 1}} x_r x_{r'} y_c y_{c'} = \mathcal{T}(g). \tag{5.5}$$

Pairs of points belonging to the same row atom make no contribution. Moreover, every value $g(x)$ is a sum of whole column-atom lengths, so the two column intervals used above never split a column atom; a pair selected from the two intervals therefore always comes from two distinct columns. Thus no row or column diagonal correction is needed. The upper bound follows from Lemma 5.1.

To see sharpness, let $R_N = C_N = [N]$ and give every vertex weight $1/N$. For $i, j \in [N]$, set

$$\sigma_N(i,j) = 0 \quad \Longleftrightarrow \quad \frac{j - 1/2}{N} \leq g_* \left(\frac{i - 1/2}{N} \right). \tag{5.6}$$

These are Ferrers matrices, and the corresponding Riemann sums converge to $\mathcal{T}(g_*) = \tau_0$. $\square$

6 Repairing partial bipartite orientations

We next prove the precise repair statement needed for the upper bound. A partial orientation of $K_{R,C}$ assigns to some pairs in $R \times C$ one of the two possible directions and leaves the remaining pairs missing. We identify state 0 with the direction $R \to C$ and state 1 with the direction $C \to R$. Thus a checkerboard of declared states is exactly a directed 4-cycle, that is, a cyclically oriented copy of $K_{2,2}$.

Lemma 6.1 (Unweighted repair lemma). *Let D be a partial orientation of $K_{R,C}$ with no directed 4-cycle, and let $m(D)$ be the number of missing pairs. There is a set $F \subseteq E(D)$ with*

$$|F| \leq m(D)$$

such that $D - F$ is acyclic.

Proof. This is Lemma 5 of Babu et al. [1]. We include its short induction because the sharp constant one is useful here. Let $\mathcal{P}(D)$ be the family of induced directed paths on four vertices. On $\mathcal{P}(D)$ introduce two equivalence relations: two paths are 2-equivalent if only their second vertices may differ, and they are 3-equivalent if only their third vertices may differ. Let $f_D(v)$ be the number of 2-classes whose paths start at v, and let $s_D(v)$ be the number of 3-classes whose paths have v as their second vertex. Reversing all arcs and reversing every path gives

$$\sum_v f_D(v) = \sum_v s_{D^{\mathrm{rev}}}(v), \qquad \sum_v s_D(v) = \sum_v f_{D^{\mathrm{rev}}}(v). \tag{6.1}$$

We induct on $|R| + |C|$. If one side of the bipartition has size at most one, then D has no directed cycle and $F = \varnothing$ works; this supplies the induction base. A source or a sink belongs to no directed cycle; we may delete it and apply induction, noting that deletion cannot increase the number of missing pairs. We may therefore suppose that every vertex has both an in-neighbour and an out-neighbour. By (6.1), after reversing every arc if necessary, there is a vertex u such that

$$f_D(u) \leq s_D(u). \tag{6.2}$$

Interchanging the two sides if necessary, assume $u \in R$. Put

$$C_1 = N^-(u), \quad C_2 = N^+(u), \quad C_3 = C \setminus (C_1 \cup C_2),$$
$$R_2 = N^+(C_2), \qquad R_1 = R \setminus (R_2 \cup \{u\}).$$

There is no arc from R_2 to C_1: such an arc, together with a two-edge path $u \to C_2 \to R_2$, would form a directed 4-cycle. Also, by the definition of R_2, there is no arc from C_2 to R_1.

Let

$$D_1 := D[R_1, C_1 \cup C_3], \qquad D_2 := D[R_2 \cup \{u\}, C_2],$$

and let E_{23} be the set of arcs directed from R_2 to C_3. For an arc $r \to c_3$ in E_{23}, choose any $c_2 \in C_2$ with $c_2 \to r$; such a choice exists by the definition of R_2. Then $u \to c_2 \to r \to c_3$ is induced: the endpoint pair u, c_3 is missing by the definition of C_3, while the other two nonconsecutive pairs u, r and c_2, c_3 lie on the same sides of the bipartition and hence are non-adjacent. All possible choices of c_2 form one 2-equivalence class, determined by (u, r, c_3). Conversely, every 2-class counted by $f_D(u)$ has this form and determines the unique arc $r \to c_3$. Therefore

$$|E_{23}| = f_D(u). \tag{6.3}$$

Likewise, for a missing pair $(c_1, r) \in C_1 \times R_2$, choosing any $c_2 \in C_2$ with $c_2 \to r$ gives the induced path $c_1 \to u \to c_2 \to r$: the endpoint pair c_1, r is missing, and the other nonconsecutive pairs c_1, c_2 and u, r are same-side pairs. Its possible choices of c_2 form one 3-equivalence class determined by (c_1, u, r), and every 3-class counted by $s_D(u)$ arises uniquely in this way. Thus those classes are in bijection with the missing pairs in $C_1 \times R_2$. Hence

$$m(D) \geq m(D_1) + m(D_2) + s_D(u). \tag{6.4}$$

By induction, choose feedback arc sets F_i in D_i with $|F_i| \leq m(D_i)$. We claim that

$$F := F_1 \cup F_2 \cup E_{23}$$

is a feedback arc set of D. Indeed, after deleting F, neither D_1 nor D_2 contains a directed cycle. A directed cycle meeting both parts would have to use an arc from D_2 to D_1. Such an arc cannot go from C_2 to R_1, cannot go from R_2 to C_1, and, if it goes from R_2 to C_3, it belongs to E_{23}. The vertex u has no arc to $C_1 \cup C_3$. Thus no such arc remains. Finally,

$$|F| \leq m(D_1) + m(D_2) + f_D(u) \leq m(D_1) + m(D_2) + s_D(u) \leq m(D),$$

as desired. Reversing all arcs does not change either the missing pairs or the minimum size of a feedback arc set, so this also covers the case in which the preliminary reversal was made. □

The lemma has the following completion form. If F is as above, take a topological order of $D - F$ and orient every missing pair, as well as every pair formerly represented by an arc of F, forwards in this order. The result is a complete acyclic orientation, and at most $m(D)$ originally specified pairs have changed direction.

The following weighted form repairs whole original cells. The random projection in its proof is important: merely projecting a clone-level feedback arc set would not give the assertion.

Corollary 6.2 (Weighted repair lemma)**.** *Let D be a partial orientation of $K_{R,C}$, with no directed 4-cycle. Give $r \in R$ weight $x_r \geq 0$ and $c \in C$ weight $y_c \geq 0$, and let*

$$M := \sum_{\substack{(r,c)\in R\times C \\ (r,c)\ \mathit{missing}}} x_r y_c.$$

There is a complete acyclic orientation of $K_{R,C}$ for which the total product weight of the originally specified pairs whose direction is changed is at most M.

Proof. First delete all vertices of weight zero and apply the argument below to the positive-weight support. Once a total order of that support has been found, reinsert the deleted vertices in arbitrary positions and orient every pair incident with a reinserted vertex forwards in the enlarged order. Any originally specified pair whose direction is thereby changed has product weight zero, so this reinsertion adds no repair cost. We may therefore assume that every weight is positive.

First suppose that all weights are rational. Choose a common denominator N and write $x_r = N_r/N$ and $y_c = N_c/N$. Replace r by N_r twins and c by N_c twins, with every clone-pair inheriting the orientation, or the missing status, of its original pair. A directed 4-cycle in the cloned orientation would project to one in D; a cycle using two twins of one vertex is impossible because twins have identical orientations. The number of missing clone-pairs is $N^2 M$.

Apply Lemma 6.1 to obtain a feedback arc set F with $|F| \leq N^2 M$, and fix a topological order $\prec$ of the cloned digraph after F is deleted. Independently and uniformly choose one clone X_v of each original vertex v. The chosen clones inherit a total order. If a specified original pair $u \to v$ is backward in this induced order, then the clone arc $X_u \to X_v$ belongs to F. Write $w_r = x_r$ for $r \in R$ and $w_c = y_c$ for $c \in C$. The expected product weight of the backward original pairs is at most

$$\begin{aligned}\sum_{u\to v} w_u w_v \Pr(X_u \to X_v \in F) &= \sum_{u\to v} \frac{|F \cap (X_u^{\mathrm{all}} \times X_v^{\mathrm{all}})|}{N^2} \\ &= \frac{|F|}{N^2} \leq M.\end{aligned}$$

Here X_u^{all} denotes the clone class of u. Fix a choice of one clone from each class for which the product weight of the backward original pairs is at most M; such a choice exists by the displayed expectation bound. Reverse precisely those backward specified pairs, and orient every missing pair forwards. Every pair is now oriented forwards in one total order, so the orientation is acyclic, and the weight of the changed specified pairs is at most M.

For arbitrary real weights, approximate them by rational weights. There are only finitely many total orders and finitely many sets of changed pairs. Passing to a subsequence on which both are fixed and then taking the limit proves the asserted closed inequality for the original weights. □

Every complete bipartite orientation without a directed 4-cycle is acyclic. Indeed, if $v_1 v_2 \dots v_{2\ell} v_1$ were a shortest directed cycle with $\ell \geq 3$, then the chord between v_1 and v_4 would, according to its direction, give either the directed 4-cycle $v_1 v_2 v_3 v_4 v_1$ or a shorter directed cycle. Its 0-neighbourhoods on one side are nested: two incomparable neighbourhoods produce one of the two checkerboards, hence a directed 4-cycle. Thus the completed matrix in Corollary 6.2 is Ferrers.

6.1 The pre-repair support reduction

We now put the preceding bipartite language into a form that applies with all quantifiers needed for the global upper bound.

Claim 6.3 (Legal marker completion and saturation). *Let $G \in \mathscr{C}$ and fix a legal partial marker map μ. There is a finite graph $\widehat{G} \in \mathscr{C}$ containing G as an ordinary subgraph and carrying a legal marker map with all q labels present, such that the following edges are present:*

1. *every all-marker edge and every neutral edge;*
2. *for every object already declaring a row or column tag, all J edges in the corresponding tag bundle;*
3. *for every object-pair already declaring state ε, all L edges in the corresponding state-ε bundle.*

Proof. For each absent label add a new isolated vertex and assign the label to it. No old vertex changes its marker/object status, so all old edges retain their profiles and all old predicates retain their truth values. Recall from the marker choice in Section 3 that $\{h\}, H \setminus \{h\}, D$, the A-block, the B-block and the two tag blocks are pairwise disjoint. In particular, $h \notin D$ and h is neither a state nor a tag marker; hence the profile types completed below cannot coincide accidentally. Now add the edges listed above. All-marker and neutral edges declare no predicate. Completing a tag bundle declares only a tag that was already declared, and completing a state bundle declares only a state that was already declared. No payload edge is added. Consequently the sets of true tag, state and payload predicates are unchanged, and none of (R1)–(R6) can become false. Thus the enlarged representation is legal. $\square$

Fix a saturated representation as in Claim 6.3 and a weight vector. Write ω_v for the weight of an object v, put $s = \sum_v \omega_v$, and define

$$r = \sum_{R_\mu(v)} \omega_v, \qquad c = \sum_{C_\mu(v)} \omega_v.$$

By (R1), $r + c \le s$. For $\varepsilon \in \{0,1\}$ let

$$D_\varepsilon = \sum_{\substack{\{u,v\} \subseteq O_\mu \\ S^\varepsilon_\mu(u,v)}} \omega_u \omega_v.$$

Rule (R2) gives

$$D_0 + D_1 \le e_2((\omega_v)_v) \le \frac{s^2}{2}.$$

Let $\mathcal{P}$ be the set of object quadruples whose payload edge is present, and set

$$A_\Sigma = \sum_i y_{A_i}, \qquad B_\Sigma = \sum_i y_{B_i}.$$

With the notation U, W, P, P' introduced later in (9.4), the exact score decomposition is

$$\begin{aligned} p_{\widehat{G}}(y,\omega) &= e_k(x) + U(rP + cP') \\ &\quad + W\left(D_0 \sum_i y_{A_i} y_{B_i} + D_1 \sum_i y_{A_i} y_{B_{i+1}}\right) + W \sum_{X \in \mathcal{P}} \prod_{v \in X} \omega_v. \end{aligned} \tag{6.5}$$

Here $x_h = y_h + s$ and $x_m = y_m$ for $m \ne h$.

The three terms following $e_k(x)$ admit bounds that do not use the repair lemma. Since the marker sets occurring in each aggregate are disjoint, weighted AM–GM gives

$$U(rP + cP') \le 2k^{-k}.$$

For the state term, use

$$\sum_i y_{A_i} y_{B_i} \le A_\Sigma B_\Sigma, \qquad \sum_i y_{A_i} y_{B_{i+1}} \le A_\Sigma B_\Sigma$$

together with $D_0 + D_1 \le s^2/2$. Hence

$$W\left(D_0 \sum_i y_{A_i} y_{B_i} + D_1 \sum_i y_{A_i} y_{B_{i+1}}\right) \\ \le \frac{1}{2} W A_\Sigma B_\Sigma s^2 \le 2k^{-k},$$

where the last inequality is weighted AM–GM with exponents $1, \dots, 1, 1, 1, 2$. Finally,

$$W \sum_{X \in \mathcal{P}} \prod_{v \in X} \omega_v \le W e_4((\omega_v)_v) \le \frac{1}{24} W s^4 < 16k^{-k}.$$

Thus, before any repair is performed, the tag, state and payload terms are bounded respectively by

$$2k^{-k}, \qquad 2k^{-k}, \qquad 16k^{-k}. \tag{6.6}$$

Lemma 6.4 (High-value support reduction)**.** *Let $k = k_\star$ be the integer certified in Section 11. Let $\widehat{G}$ be saturated as above, and choose an optimal weighting whose positive support has minimum cardinality. If*

$$\lambda(\widehat{G}) > b_q + 10q^{-k},$$

then

$$|qx_m - 1| \le r_k \quad (m \in \mathcal{M}), \qquad r_k := q\sqrt{\frac{30k^{-k}}{b_q}}, \tag{6.7}$$

$y_h > 0$, every positive-weight object has exactly one of the row and column tags, every positive-weight stated pair is a row–column pair, and every positive-weight payload has two row and two column objects.

Proof. By Lemma 4.2, the positive support is pair-covered. The value is larger than b_q, and (6.5)–(6.6) show that

$$b_q - e_k(x) < 30k^{-k}.$$

Maclaurin's inequality gives

$$\frac{e_k(x)}{b_q} \le \left(1 - \frac{q}{q-1} \sum_{m \in \mathcal{M}} (x_m - q^{-1})^2\right)^{k/2}.$$

Since $1 - (1-z)^{k/2} \ge z$ for $0 \le z \le 1$, this implies (6.7). In particular, all x_m are positive for the fixed integer of Section 11.

Suppose that $y_h = 0$. Since $h \in H$, the tag term vanishes, and $s = x_h$. In the tube (6.7),

$$W\left(D_0 \sum_i y_{A_i} y_{B_i} + D_1 \sum_i y_{A_i} y_{B_{i+1}}\right) \le \frac{L}{2}(1 + r_k)^k q^{-k},$$

while

$$W \sum_{X \in \mathcal{P}} \prod_{v \in X} \omega_v \le \frac{1}{24}(1 + r_k)^k q^{-k}.$$

Therefore

$$p_{\widehat{G}}(y,\omega) \le b_q + \left(\frac{L}{2} + \frac{1}{24}\right)(1+r_k)^k q^{-k}. \tag{6.8}$$

Condition (11.5) makes the second term strictly smaller than $10q^{-k}$, contrary to the hypothesis. Hence $y_h > 0$.

The marker h and every positive-weight object now form a pair in the positive support, so pair-cover supplies an edge containing them. By the profile list, the only edges containing both h and an object are row- or column-tag edges: neutral profiles omit h, state and payload profiles use the disjoint marker set D, and all-marker edges contain no object. Thus every positive-weight object is tagged, and (R1) makes its tag unique. Rules (R3) and (R5) now give the asserted forms of positive stated pairs and payloads. Notice that no conclusion is asserted here for zero-weight objects; they will be deleted, without changing the weighted score, before the repair step in Lemma 12.1. □

There is a fixed legal tagged test template that places the high-value branch strictly above the threshold in the preceding lemma. Take one row object and one column object, insert all their tag bundles, give their pair state 0, and insert every all-marker and neutral edge. With

$$y_h = s = \frac{1}{2q}, \qquad a = b = \frac{1}{4q}, \qquad y_m = \frac{1}{q} \quad (m \ne h),$$

the base value is b_q, while the tag and state terms are exactly

$$\left(\frac{J}{4} + \frac{L}{16}\right) q^{-k} = \frac{101}{4} q^{-k} = 25.25 q^{-k}. \tag{6.9}$$

Consequently

$$\Lambda(\mathscr{C}) \ge b_q + \frac{101}{4} q^{-k} > b_q + 10q^{-k}.$$

7 The completion inequality

We isolate the score comparison used in the proof of the structural theorem. This also makes clear why a merely qualitative removal lemma would not suffice.

Let R, C be the row and column objects, of total weights a, b. A specified cell has state 0 or state 1 and contributes, respectively, $z_0 x_r y_c$ or $z_1 x_r y_c$ to the state part of the score. A payload has two row objects and two column objects and contributes $W x_r x_{r'} y_c y_{c'}$. In a partial matrix it may be present even when some of its four cells are missing; if all four cells are specified, the local rule requires that exactly one of them have state 1. Put

$$z_{\min} := \min\{z_0, z_1\}, \qquad \Delta := |z_0 - z_1|.$$

Lemma 7.1 (Score-preserving completion)**.** *Suppose that the specified cells contain no directed 4-cycle and that*

$$z_{\min} \ge \Delta + 2Wab. \tag{7.1}$$

The partial state matrix can be replaced by a complete Ferrers matrix without decreasing the total state-plus-payload score. Once the matrix is complete, adding all permitted payloads can only increase the score, and their total weight is at most

$$\tau_0 W a^2 b^2. \tag{7.2}$$

Proof. Let M be the product weight of the missing cells. By Corollary 6.2, we may complete the matrix after changing specified cells of total weight $R_0 \le M$. Filling the missing cells gains at least $z_{\min} M$. Changing a specified cell loses at most Δ times its product weight in the state score.

Delete every old payload that contains a missing or a changed cell. For a fixed cell (r, c), the total product weight of quadruples containing it is at most $x_r y_c ab$. By the union bound, the deleted payload score is at most $Wab(M + R_0)$. Every old payload not deleted is still permitted after the repair. The change in score is therefore at least

$$z_{\min} M - \Delta R_0 - Wab(M + R_0) \ge \big(z_{\min} - \Delta - 2Wab\big) M \ge 0.$$

The resulting complete acyclic orientation is Ferrers. We may now add all permitted payloads and apply Corollary 5.2, which gives (7.2). $\square$

Lemma 7.1 is a comparison of weighted state matrices. When it is applied to an actual weighted template, the completed matrix is used only to dominate the numerical score; no assertion is made that all of its repaired states or newly permitted payloads occur as edges of the same finite hypergraph.

In the marker construction, if A_i, B_i are the marker weights occurring in the two cyclic state bundles and W is the product of the common payload markers, then

$$z_0 = W \sum_{i \in \mathbb{Z}/L\mathbb{Z}} A_i B_i, \qquad z_1 = W \sum_{i \in \mathbb{Z}/L\mathbb{Z}} A_i B_{i+1}. \tag{7.3}$$

By Lemma 6.4, every high-value weighting in the upper-bound sequence obeys (6.7); for the fixed k we have $r_k < 1/10$, and hence

$$\frac{9}{10q} \le A_i, B_i \le \frac{11}{10q}, \qquad a + b \le x_h \le \frac{11}{10q}. \tag{7.4}$$

This pre-repair localization avoids any circular appeal to the completed functional.

With $L = 4$, (7.4) gives throughout the only region in which a global maximum can occur

$$z_{\min} \ge \Delta + 2Wab, \qquad W > 0 \Longrightarrow z_{\min} > \Delta + 2Wab. \tag{7.5}$$

Here is an explicit numerical check of the margin. It is enough that

$$\frac{9}{10q} \le A_i, B_i \le \frac{11}{10q} \quad (i \in \mathbb{Z}/4\mathbb{Z}), \qquad a + b \le \frac{11}{10q}, \tag{7.6}$$

and (7.4) implies this rational box. If $W = 0$, then $z_0 = z_1 = 0$ and the score comparison is trivial. We may therefore assume $W > 0$ and divide by W;

$$\frac{z_{\min}}{W} \ge \frac{4(9/10)^2}{q^2} = \frac{81}{25q^2}, \qquad \frac{\Delta}{W} \le \sum_i A_i |B_i - B_{i+1}| \le \frac{22}{25q^2},$$

whereas

$$2ab \le \frac{(a+b)^2}{2} \le \frac{121}{200q^2}.$$

Thus the right-hand side of (7.5), divided by W, is at most $297/(200q^2) < 81/(25q^2)$. Thus (7.1) holds, even before exact cyclic symmetrization has established $z_0 = z_1$. This observation avoids any circular use of the symmetry conclusion in the repair step. After symmetrization, $A_i = B_i = A$ and hence

$$z_0 = z_1 = LWA^2. \tag{7.7}$$

Combining Lemma 7.1, Corollary 5.2, and the marker symmetrization lemma replaces the entire object contribution by its upper bound

$$LWA^2ab + \tau_0 W a^2 b^2. \tag{7.8}$$

In the final symmetry class $a = b = s/2$, this is

$$\frac{L}{4} W A^2 s^2 + \frac{\tau_0}{16} W s^4. \tag{7.9}$$

This is exactly the last two terms of the rational polynomial $\Phi_k(\tau_0)$ defined in the structural theorem. Hence the repair loses neither the exact coefficient nor the strict margin needed for the global upper bound.

8 Finite lower-bound approximants

We next verify that the Ferrers coefficient is obtained from finite admissible templates, rather than only from a graphon. For $N \geq 1$, let $R_N = \{r_1, \dots, r_N\}$ and $C_N = \{c_1, \dots, c_N\}$, and define the states by (5.6). Declare every cell, and insert a payload on $\{r_i, r_{i'}, c_j, c_{j'}\}$ if and only if its four cells contain exactly one entry of state 1. The matrix is Ferrers, so all the local no-checkerboard rules are satisfied.

Let $d_i := |N_0(r_i)|$. Since $d_1 \geq \cdots \geq d_N$, the number P_N of payloads is exactly

$$|P_N| = \sum_{1 \leq i < i' \leq N} d_{i'}(d_i - d_{i'}). \tag{8.1}$$

It follows either directly from (8.1), or from the Riemann sum in Corollary 5.2, that

$$\lim_{N \to \infty} \frac{|P_N|}{N^4} = \tau_0. \tag{8.2}$$

Fix a maximizing point (y, s) in the compact optimization defining $\Phi_k(\tau_0)$, set $a = b = s/2$, and give each row object weight a/N and each column object weight b/N. Insert every all-marker edge of profile (P0) and every neutral edge of profile (PN). For each row object insert exactly its full row-tag bundle (PR), and for each column object exactly its full column-tag bundle (PC). For every cross-pair insert exactly the full state bundle (PS0) or (PS1) prescribed by the matrix, and insert a payload edge (PP) exactly for the quadruples in P_N. No other tag, state or payload edge is inserted. Denote the resulting finite template by G_N. The local rules defining the marker class are certificate-positive and are satisfied by this construction: the tags are disjoint, every cross-pair has one state, the matrix has no checkerboard, and every payload has exactly one state-1 cell. Therefore $G_N \in \mathscr{C}$.

The state contribution is exactly LWA^2ab, and by (8.2) the payload contribution converges to $\tau_0 W a^2 b^2$. All other terms are independent of N. Consequently

$$\lim_{N \to \infty} p_{G_N}(\boldsymbol{x}_N) = \Phi_k(\tau_0), \tag{8.3}$$

where $p_G(\boldsymbol{x}) := \sum_{e\in E(G)} \prod_{v\in e} x_v$ denotes the unscaled edge polynomial. Lagrangian weights are allowed to be real, so no rational approximation of the maximizing marker vector is needed. Since $\lambda(G_N) \ge p_{G_N}(\boldsymbol{x}_N)$, (8.3) gives the lower bound

$$\Lambda(\mathscr{C}) \ge \Phi_k(\tau_0). \tag{8.4}$$

The matching upper bound, with all supremum quantifiers made explicit, is proved in Proposition 12.2 after the marker localization and exact symmetrization have been established.

9 Localization of the marker weights

We keep the notation of the preceding sections. Thus $L = 4$, $J = 100$, k is an integer to be fixed below, and

$$q = k^3.$$

The marker set is denoted by $\mathcal{M}$, $|\mathcal{M}| = q$. The weight of the marker m is y_m. The total weights of the row and column objects are a and b, respectively, and

$$s = a + b.$$

All weights are non-negative and

$$\sum_{m\in\mathcal{M}} y_m + s = 1. \tag{9.1}$$

Set

$$x_h = y_h + s, \qquad x_m = y_m \quad (m \ne h). \tag{9.2}$$

Then $x = (x_m)_{m\in\mathcal{M}}$ belongs to the standard $(q-1)$-simplex. The all-marker edges and the saturated neutral edges contribute

$$e_k(y) + s e_{k-1}(y_m : m \ne h) = e_k(x). \tag{9.3}$$

This identity is the reason for singling out the marker h.

We shall use the abbreviations

$$\begin{aligned} U &= \prod_{m\in H} y_m, & W &= \prod_{m\in D} y_m, \\ P &= \sum_{j=1}^{J} y_{p_j}, & P' &= \sum_{j=1}^{J} y_{p'_j}, \\ Z_0 &= W \sum_{i\in\mathbb{Z}/L\mathbb{Z}} y_{A_i} y_{B_i}, & Z_1 &= W \sum_{i\in\mathbb{Z}/L\mathbb{Z}} y_{A_i} y_{B_{i+1}}. \end{aligned} \tag{9.4}$$

After applying the weighted repair lemma, let E_0 and E_1 denote the total product weights of the cross-pairs in states 0 and 1. The repaired code is complete, and hence

$$E_0, E_1 \ge 0, \qquad E_0 + E_1 = ab. \tag{9.5}$$

Let $\mathcal{D}$ be the set of all nonnegative tuples

$$(y, a, b, E_0, E_1)$$

satisfying

$$\sum_{m\in\mathcal{M}} y_m + a + b = 1, \qquad E_0 + E_1 = ab.$$

This is a compact semialgebraic set over $\mathbb{Q}$. The domain $\mathcal{D}$ is deliberately a *numerical relaxation*: every numerical repair performed on the positive support of an actual finite weighted template supplies a point of $\mathcal{D}$, but a general point of $\mathcal{D}$ need not be realized by any graph, let alone by the graph from which its marker weights arose. Only the forward implication is used for the upper bound. For $0 \le t \le 1$ define

$$\mathcal{Q}_t(y, a, b, E_0, E_1) = e_k(x) + U(aP + bP') + E_0 Z_0 + E_1 Z_1 + tWa^2b^2, \tag{9.6}$$

where $s = a + b$, $x_h = y_h + s$, and $x_m = y_m$ for $m \neq h$. Write

$$\Psi_k(t) := \max_{\mathcal{D}} \mathcal{Q}_t. \tag{9.7}$$

At the value used in the construction, $t = \tau_0 = 1/(2e^2)$, the last term is precisely the weighted Ferrers upper bound. It is useful that every estimate in this section is uniform on the larger rational interval $0 \le t \le 1$.

Put

$$b_q = \binom{q}{k} q^{-k}, \qquad c_k = \binom{q-2}{k-2} q^{-(k-2)}. \tag{9.8}$$

Thus $b_q = e_k(q^{-1}, \ldots, q^{-1})$, while $-c_k$ is the eigenvalue of the Hessian of e_k at the uniform point on the tangent space of the simplex.

Lemma 9.1 (Uniform global localization)**.** *There is an absolute integer k_0 such that the following holds for every $k \ge k_0$ and every $t \in [0, 1]$. If a weighting maximizes (9.6), then*

$$V(x) := \sum_{m \in \mathcal{M}} (x_m - q^{-1})^2 \le v_k := \frac{30k^{-k}}{b_q}. \tag{9.9}$$

In particular, with

$$r_k = q\sqrt{v_k}, \tag{9.10}$$

we have

$$|qx_m - 1| \le r_k \quad (m \in \mathcal{M}), \qquad r_k \longrightarrow 0. \tag{9.11}$$

Moreover, no x_m is zero.

Proof. Every term after $e_k(x)$ in (9.6) is non-negative. We first give a global upper bound for their sum. The weighted AM–GM inequality, applied to the k factors in each profile, gives

$$U(aP + bP') \le 2k^{-k}, \qquad E_0 Z_0 + E_1 Z_1 \le 2Lk^{-k}, \qquad Wa^2b^2 \le 16k^{-k}.$$

For the state term, $0 \le E_\varepsilon \le ab$ on $\mathcal{D}$; hence each summand is bounded by a product having the k factors consisting of the $k - 4$ coordinates indexed by D, one A-coordinate, one B-coordinate, a, and b. In the payload term the optimal AM–GM allocation gives weight $2/k$ to each of a, b and $1/k$ to each marker in D, whence the factor 16. Since $L = 4$, the total is at most $26k^{-k} < 30k^{-k}$.

For x in the simplex, Maclaurin's inequality gives

$$\frac{e_k(x)}{b_q} \le \left(1 - \frac{qV(x)}{q-1}\right)^{k/2}. \tag{9.12}$$

Indeed,

$$e_2(x) = \frac{1}{2}\left(1 - \sum_m x_m^2\right) = \frac{q-1}{2q} - \frac{V(x)}{2},$$

and (9.12) is the comparison of the second and the kth elementary symmetric means. The point of $\mathcal{D}$ with $s = a = 1/(2q)$, $b = E_0 = E_1 = 0$, $y_h = 1/(2q)$, and $y_m = 1/q$ for $m \neq h$ has value at least b_q. Hence a maximizing weighting satisfies

$$b_q - e_k(x) \leq 30k^{-k}.$$

Since $1 - (1-z)^{k/2} \geq z$ for $0 \leq z \leq 1$ and $k \geq 2$, (9.9) follows from (9.12).

If some x_m were zero, let $r \leq q-1$ be the size of its support. Maclaurin's inequality on that support gives $e_k(x) \leq b_r$, where

$$b_r := \binom{r}{k} r^{-k} = \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{r}\right) \qquad (r \geq k).$$

Each factor in the product is non-decreasing in r, so b_r is non-decreasing for $r \geq k$; if $r < k$, then $e_k(x) = 0$. Consequently

$$e_k(x) \leq b_{q-1}, \qquad b_{q-1} := \binom{q-1}{k} (q-1)^{-k}.$$

For $q = k^3$ and all sufficiently large k,

$$b_q - b_{q-1} \geq \frac{k(k-1)}{4k!q^2} > 30k^{-k}. \tag{9.13}$$

To verify the first inequality, put

$$a_j = 1 - \frac{j}{q}, \qquad d_j = 1 - \frac{j}{q-1} \qquad (0 \leq j \leq k-1).$$

Then $a_j \geq d_j$ and the exact telescoping identity

$$\prod_{j=0}^{k-1} a_j - \prod_{j=0}^{k-1} d_j = \sum_{i=0}^{k-1} (a_i - d_i) \left(\prod_{j<i} d_j\right) \left(\prod_{j>i} a_j\right)$$

holds. Every mixed product on the right is at least $1/2$ once $q = k^3$ and k is large, while

$$\sum_{i=0}^{k-1} (a_i - d_i) = \sum_{i=0}^{k-1} \left(\frac{i}{q-1} - \frac{i}{q}\right) = \frac{k(k-1)}{2q(q-1)}.$$

After division by $k!$, this gives

$$b_q - b_{q-1} \geq \frac{k(k-1)}{4k!q(q-1)} \geq \frac{k(k-1)}{4k!q^2}.$$

For the second inequality in (9.13), observe that

$$\frac{k!}{k^k} \leq 2^{-\lfloor k/2 \rfloor},$$

because the first $\lfloor k/2 \rfloor$ factors in $k!/k^k = \prod_{i=1}^{k} (i/k)$ are at most $1/2$. Since $q = k^3$, after clearing denominators it is enough that $120k^6(k!/k^k) < k(k-1)$, which holds for all sufficiently large k. The test weighting above now contradicts maximality.

Finally, the elementary product bound $\prod_j (1-u_j) \ge 1 - \sum_j u_j$ for $u_j \in [0,1]$ gives

$$b_q = \frac{1}{k!}\prod_{j=0}^{k-1}\left(1-\frac{j}{q}\right) \ge \frac{1}{k!}\left(1-\frac{k(k-1)}{2q}\right) \ge \frac{1}{2k!}$$

for all sufficiently large k. Consequently

$$q^2 v_k \le 60 q^2 \frac{k!}{k^k} \le 60 k^6 2^{-\lfloor k/2 \rfloor} \longrightarrow 0,$$

which proves (9.11). □

The next lemma separates the one flat direction of (9.3) from all transverse directions. Its quantitative formulation will also make clear that one integer k works simultaneously for every $t \in [0,1]$.

We shall repeatedly use the following elementary estimate. If $M(z) = C\prod_i z_i^{d_i}$ has total degree k, $|C| \le C_*$, and at most k of the exponents d_i are non-zero, then, on the box $0 \le z_i \le (1+r)/q$,

$$\max_i |\partial_i M| \le C_* k(1+r)^k q^{-(k-1)}, \qquad \|\nabla^2 M\|_{\mathrm{op}} \le C_* k^3 (1+r)^k q^{-(k-2)}. \tag{9.14}$$

Indeed, the coefficient produced by one differentiation is at most k, the coefficient produced by two is at most k^2, and a Hessian row has at most k non-zero entries. Aggregates such as P are first expanded into their J monomials. Since $J = 100$ and $L = 4$ are fixed, the sum of all constants C_* occurring below is less than 10^5.

Lemma 9.2 (Transverse curvature)**.** *For all sufficiently large k, throughout the tube $V(x) \le v_k$, and for every vector ξ with $\sum_m \xi_m = 0$, one has*

$$\xi^{\mathsf{T}} \nabla^2 e_k(x) \xi \le -\frac{c_k}{2}\|\xi\|_2^2. \tag{9.15}$$

The Hessian, in the marker variables, of the part of (9.6) following $e_k(x)$ has operator norm at most

$$B_k := 10^8 k^4 q^{-(k-2)}, \tag{9.16}$$

and $B_k < c_k/4$.

Proof. At the uniform point $\boldsymbol{u} = (1/q, \dots, 1/q)$, the diagonal entries of $\nabla^2 e_k$ vanish and every off-diagonal entry equals c_k. Thus its restriction to $\sum_m \xi_m = 0$ is exactly $-c_k I$.

For completeness, we estimate the perturbation. By the mean value theorem, each Hessian entry changes by at most

$$\sqrt{q v_k}\binom{q-3}{k-3}\left(\frac{1+r_k}{q}\right)^{k-3}.$$

Taking row sums and dividing by c_k shows that the operator-norm change is at most

$$2kq^{3/2}\sqrt{v_k}(1+r_k)^k\, c_k. \tag{9.17}$$

The coefficient in (9.17) tends to zero exponentially, by (9.9); this proves (9.15).

We now count the perturbative monomials explicitly. After expanding P and P', the tag term consists of $2J$ monomials. The state term consists of $2L$ marker monomials multiplied by E_0 or E_1; on $\mathcal{D}$,

$$0 \le E_\varepsilon \le ab \le s^2/4,$$

so each is dominated, together with its first two marker derivatives, by a degree-k monomial having the two aggregate factors a, b. The payload term contributes one further monomial. Thus there are exactly $2J + 2L + 1 = 209$ relevant monomial bounds. In the tube every marker factor, and also s because $s \le x_h$, is at most $(1 + r_k)/q$. Applying (9.14) term by term and taking row sums gives

$$\|\nabla^2(\mathcal{Q}_t - e_k)\|_{\mathrm{op}} \le 10^6 k^3 (1 + r_k)^k q^{-(k-2)} \le B_k.$$

Finally

$$c_k \ge \frac{1}{2(k-2)!}$$

for all sufficiently large k, whereas $k^4 (k-2)! q^{-(k-2)} \to 0$. This proves $B_k < c_k/4$. $\square$

10 Exact symmetrization

We next justify the equalities among marker and object weights. They are exact equalities, not limiting assertions.

Lemma 10.1 (Exact marker and row–column symmetry)**.** *For all sufficiently large k, uniformly for $t \in [0, 1]$, every maximizer of $\mathcal{Q}_t$ on $\mathcal{D}$ satisfies*

$$\begin{gathered} y_{p_1} = \cdots = y_{p_J} = y_{p'_1} = \cdots = y_{p'_J} =: p, \\ y_{A_i} = y_{B_i} =: A \quad (i \in \mathbb{Z}/L\mathbb{Z}), \\ a = b = s/2. \end{gathered} \tag{10.4}$$

In particular $Z_0 = Z_1 = LWA^2$.

Proof. The coordinates within each of the two p-groups are equal. Indeed, all terms other than $e_k(x)$ depend on such a group only through its sum. If two coordinates u, v in the group have fixed sum, then

$$e_k(x) = C_0 + C_1(u + v) + C_2 uv,$$

where $C_2 = e_{k-2}$ of the remaining coordinates is positive by Lemma 9.1. Replacing u, v by their average therefore strictly increases the value unless $u = v$.

We next equalize the A- and B-groups. Write

$$y_{A_i} = \bar{A} + \alpha_i, \qquad y_{B_i} = \bar{B} + \beta_i, \qquad \sum_i \alpha_i = \sum_i \beta_i = 0.$$

Let $\widehat{y}$ be obtained from y by replacing the A_i-coordinates by $\bar{A}$ and the B_i-coordinates by $\bar{B}$, let $\widehat{x}$ be the associated vector from (9.2), and put $\delta = x - \widehat{x}$. This averaging is the orthogonal projection of those two coordinate blocks onto their constant subspaces. It fixes the uniform point and therefore satisfies $V(\widehat{x}) \le V(x) \le v_k$. Since the tube is the intersection of the simplex with a Euclidean ball centered at the uniform point, it is convex; hence the entire segment from $\widehat{x}$ to x lies in the tube.

Suppress the fixed variables a, b, E_0, E_1 and the marker coordinates outside the two blocks. At $\widehat{x}$, the marker gradient of the full functional $\mathcal{Q}_t$ is constant within each A-block and within each B-block. For e_k this follows from permutation symmetry. For the only other term involving these coordinates, one has at $\widehat{x}$

$$\partial_{y_{A_i}}(E_0 Z_0 + E_1 Z_1) = W(E_0 + E_1)\bar{B}, \qquad \partial_{y_{B_i}}(E_0 Z_0 + E_1 Z_1) = W(E_0 + E_1)\bar{A},$$

independently of i. Identifying δ with its corresponding marker-coordinate vector (zero outside the A- and B-blocks), we obtain $\nabla_y \mathcal{Q}_t(\widehat{y}, a, b, E_0, E_1) \cdot \delta = 0$. By Lemma 9.2, along the segment and for every simplex-tangent marker vector ξ,

$$\xi^{\mathsf{T}} \nabla_y^2 \mathcal{Q}_t \, \xi \leq -\left(\frac{c_k}{2} - B_k\right) \|\xi\|_2^2 \leq -\frac{c_k}{4} \|\xi\|_2^2. \tag{10.5}$$

Taylor's formula with integral remainder now gives

$$\mathcal{Q}_t(\widehat{y}, a, b, E_0, E_1) - \mathcal{Q}_t(y, a, b, E_0, E_1) \geq \frac{c_k}{8} \|\delta\|_2^2 = \frac{c_k}{8} \left(\sum_i \alpha_i^2 + \sum_i \beta_i^2 \right). \tag{10.6}$$

Thus maximality forces all α_i and β_i to vanish.

It remains within these groups to show $\bar{A} = \bar{B}$. Put $m = (\bar{A} + \bar{B})/2$ and $\gamma = (\bar{A} - \bar{B})/2$, and replace all $2L$ coordinates by m. This is again an orthogonal projection toward the uniform point, so both the averaged point and the joining segment remain in the localization tube. At the new point all their first derivatives of e_k agree, while the deviation has L entries γ and L entries $-\gamma$. The same Taylor argument gives a base gain at least $c_k L \gamma^2/2$. The state term changes from $LWab(m^2 - \gamma^2)$ to $LWabm^2$, and hence also gains $LWab\gamma^2$. Thus $\gamma = 0$ at a maximizer, and $y_{A_i} = y_{B_i} = A$ for every i.

Finally couple the two tag groups with the two object classes. Write

$$y_{p_j} = p + \varepsilon, \qquad y_{p'_j} = p - \varepsilon, \qquad a = s/2 + d, \qquad b = s/2 - d.$$

Replacing these quantities by $p, p, s/2, s/2$ preserves total weight and x_h. At the same time replace (E_0, E_1) by any nonnegative pair with sum $s^2/4$ (for example, $(s^2/4, 0)$); the resulting tuple still belongs to $\mathcal{D}$. This redistribution is made only in the numerical relaxation, and no finite template realizing the new split is asserted. The replacement is an orthogonal projection of the p-coordinates onto their common-mean subspace. It fixes the uniform point, hence $V(x^*) \leq V(x)$; convexity of the tube then places the entire segment from x^* to x inside it. At x^* all $2J$ p-coordinates have the same first derivative of e_k, so the deviation, consisting of J entries ε and J entries $-\varepsilon$, is orthogonal to that gradient. Taylor's formula therefore yields

$$e_k(x^*) - e_k(x) \geq \frac{c_k J}{2} \varepsilon^2.$$

The original tag term exceeds the symmetric one by exactly

$$2JUd\varepsilon. \tag{10.7}$$

Since $Z_0 = Z_1 =: z = LWA^2$, the symmetric state term exceeds the original one by exactly zd^2. The payload gain is

$$tW \left(\frac{s^4}{16} - \left(\frac{s^2}{4} - d^2 \right)^2 \right) = tW \left(\frac{s^2 d^2}{2} - d^4 \right) \geq 0.$$

Consequently the total gain from symmetrizing is at least the quadratic form

$$\frac{c_k J}{2} \varepsilon^2 + zd^2 - 2JUd\varepsilon = \begin{pmatrix} \varepsilon & d \end{pmatrix} \begin{pmatrix} c_k J/2 & -JU \\ -JU & z \end{pmatrix} \begin{pmatrix} \varepsilon \\ d \end{pmatrix}.$$

Its leading diagonal entry is positive, and the determinant is

$$\det = \frac{J}{2} \bigl(c_k z - 2JU^2 \bigr).$$

Thus the form is positive definite, and equality forces $d = \varepsilon = 0$, provided

$$2JU^2 < c_k z. \tag{10.8}$$

In the localized tube,

$$\frac{2JU^2}{c_k z} \le \frac{4J}{L}(k-2)!q^{-(k-2)}\frac{(1+r_k)^{2k}}{(1-r_k)^k} = o(1), \tag{10.9}$$

and Section 11 verifies the strict inequality for $k = k_\star$. Hence $d = \varepsilon = 0$, proving (10.4). $\square$

After Lemma 10.1, let $\mathcal{K}$ be the compact rational simplex cut out by (9.1), nonnegativity, and the rational linear equalities in (10.4). On this simplex consider the polynomial

$$\begin{aligned} F_{k,t}(y,s) = e_k(y) + se_{k-1}(y_m : m \neq h) + JUsp \\ + \frac{L}{4}WA^2s^2 + \frac{t}{16}Ws^4. \end{aligned} \tag{10.10}$$

We denote its maximum by

$$\Phi_k(t) := \max_{(y,s)\in\mathcal{K}} F_{k,t}(y,s). \tag{10.11}$$

Corollary 10.2 (Equality of the enlarged and symmetric maxima)**.** *For every sufficiently large k and every $t \in [0,1]$,*

$$\Psi_k(t) = \Phi_k(t).$$

Proof. The compact set $\mathcal{D}$ has a maximizer of $\mathcal{Q}_t$. Lemma 10.1 shows that every such maximizer satisfies (10.4); at that point $Z_0 = Z_1 = LWA^2$, and $E_0 + E_1 = ab = s^2/4$, so its value is exactly $F_{k,t}(y,s)$. Hence $\Psi_k(t) \le \Phi_k(t)$. Conversely, every point of $\mathcal{K}$ lifts numerically to a point of $\mathcal{D}$ by taking $a = b = s/2$ and, for instance, $E_0 = ab$, $E_1 = 0$. No graph realization of this relaxed lift is asserted or needed. At the lift $\mathcal{Q}_t = F_{k,t}$, proving the reverse inequality. $\square$

11 Fixing the uniformity

We now freeze all parameters. Recall that $L = 4$, $J = 100$, $q = k^3$,

$$b_q = \binom{q}{k}q^{-k}, \qquad c_k = \binom{q-2}{k-2}q^{-(k-2)}, \qquad v_k = \frac{30k^{-k}}{b_q}.$$

We take

$$k_\star := 1000, \qquad q := k_\star^3 = 10^9.$$

The reserved markers fit in this value of q, and the following finite list of rational inequalities holds (with $k = k_\star$). Their exact integer cross-multiplications and a machine-readable certificate are given in Appendix A:

$$30k^{-k} < \frac{k(k-1)}{4k!q^2}, \tag{11.1}$$

$$q^2 v_k < \frac{1}{10^{12}k^2q^2}, \tag{11.2}$$

$$10^8k^4q^{-(k-2)} < \frac{c_k}{4}, \tag{11.3}$$

$$\frac{16J}{L}(k-2)!q^{-(k-2)} < 1, \tag{11.4}$$

$$\left(\frac{L}{2}+\frac{1}{24}\right)\left(1+\frac{1}{10^6k}\right)^k<10. \tag{11.5}$$

Thus the choice of uniformity is completely explicit.

Lemma 11.1. *With $k=k_\star=1000$, all localization, repair, curvature and symmetrization estimates in Section 6.1 and from Section 9 through Section 10 hold simultaneously, uniformly for $t\in[0,1]$.*

Proof. The number of reserved labels is $2k-6+2L+2J<q$. Also

$$b_q=\frac{1}{k!}\prod_{i=0}^{k-1}\left(1-\frac{i}{q}\right)\geq\frac{1}{2k!}$$

because the sum of the numbers i/q is smaller than $1/2$. Moreover, the same product estimate gives $c_k\geq 1/(2(k-2)!)$. Also

$$\frac{k!}{k^k}\leq 2^{-\lfloor k/2\rfloor},$$

by bounding the first $\lfloor k/2\rfloor$ factors of $k!/k^k$ by $1/2$. Hence

$$q^2v_k\leq 60q^2\frac{k!}{k^k}\leq 60k^62^{-500}.$$

Since $2^{500}>10^{150}$ and $q=10^9$, the preceding estimate gives

$$q^2v_k<60\cdot10^{18}\cdot10^{-150}=6\cdot10^{-131}<10^{-36}=\frac{1}{10^{12}k^2q^2}.$$

This proves (11.2) at $k=1000$ without a hidden order-of-magnitude step. For the product-gap estimate (9.13), every mixed product in its telescoping identity is at least

$$\prod_{j=0}^{k-1}\left(1-\frac{j}{q-1}\right)\geq 1-\frac{k(k-1)}{2(q-1)}>\frac{1}{2}.$$

Thus (11.1) reduces to

$$120k^6\frac{k!}{k^k}<k(k-1),$$

which follows from $k!/k^k\leq 2^{-500}$; for instance $121\cdot10^{12}<2^{47}<2^{500}$.

For (11.3), after cancelling $q^{-(k-2)}$ it is enough to show

$$4\cdot10^8k^4<\binom{q-2}{k-2}.$$

The left side is $4\cdot10^{20}$, while monotonicity of the binomial coefficients up to their middle gives

$$\binom{q-2}{k-2}\geq\binom{q-2}{3}>10^{26}.$$

Next, $998!<1000^{998}$ gives

$$\frac{16J}{L}(k-2)!<10^{2997}<10^{8982}=q^{k-2},$$

which is (11.4). Finally, the binomial theorem gives, for $0 \le kx < 1$,

$$(1+x)^k \le \sum_{j\ge 0}(kx)^j = \frac{1}{1-kx}.$$

With $x = 10^{-9}$, the last quantity is smaller than 2; since $2(L/2 + 1/24) = 49/12 < 10$, this proves (11.5).

By (11.2), the localization radius satisfies

$$r_k = q\sqrt{v_k} < \frac{1}{10^6 kq}.$$

In particular,

$$2kq^{3/2}\sqrt{v_k}(1+r_k)^k = 2k\sqrt{q}\, r_k(1+r_k)^k < \frac{4}{10^6\sqrt{q}} < \frac{1}{2},$$

where $(1+r_k)^k < 2$ follows from the same geometric-series estimate. Thus the Hessian perturbation in (9.17) is covered without an additional asymptotic threshold. The same bound also gives

$$10^6 k^3(1+r_k)^k q^{-(k-2)} < 10^8 k^4 q^{-(k-2)} = B_k,$$

so (11.3) applies to the full perturbative Hessian estimate. All non-h marker coordinates lie in $[(1-r_k)/q, (1+r_k)/q]$, and since $s \le x_h$, the repair box (7.4) holds. Moreover, the same elementary binomial and Bernoulli estimates give the explicit bounds

$$(1+r_k)^k \le \frac{1}{1-kr_k} < 2, \qquad \frac{(1+r_k)^{2k}}{(1-r_k)^k} \le \frac{1}{(1-2kr_k)(1-kr_k)} < 4.$$

Condition (11.3) is exactly the full-Hessian domination used in the A- and B-block averaging in (10.5). For the remaining determinant condition in Lemma 10.1, the tube and $c_k \ge 1/(2(k-2)!)$ give

$$\frac{2JU^2}{c_k LWA^2} \le \frac{4J}{L}(k-2)!q^{-(k-2)}\frac{(1+r_k)^{2k}}{(1-r_k)^k} < 1,$$

by (11.4) and the second bound in the preceding display. Thus $2JU^2 < c_k LWA^2$ holds with strict margin. Condition (11.5) makes the $y_h = 0$ upper bound strictly smaller than the $25.25q^{-k}$ tagged test. Hence one and the same integer works in every preceding estimate and for every $t \in [0,1]$. $\square$

From now on

$$k := k_\star, \qquad q := k^3. \tag{11.6}$$

In particular, the uniformity has been fixed before any limit in the order of a host hypergraph is taken. Its definition uses neither e nor any real input.

Dependency diagram

The proof uses three logically different passages: finite forcing and ordinary containment, numerical domination of a weighted template, and a separate finite lower approximation. The following diagram records their directions; no reverse arrow from $\mathcal{D}$ to an actual graph is used.

Finite forcing and transfer: profiles $\Longrightarrow \mathscr{C} = \mathrm{Forb}(\mathcal{A}) \Longrightarrow \mathcal{F} \Longrightarrow \pi(\mathcal{F}) = k!\Lambda(\mathscr{C})$.

Upper domination: $G^{(\ell)} \in \mathscr{C} \Longrightarrow \widehat{G}^{(\ell)} \Longrightarrow (\widehat{G}^{(\ell)}, \boldsymbol{\omega}^{(\ell)}) \Longrightarrow (\widehat{G}^{(\ell)})^+$
$\Longrightarrow$ positive-support partial matrix $\Longrightarrow \xi_\ell \in \mathcal{D} \Longrightarrow \mathcal{Q}_{\tau_0}(\xi_\ell) \le \Phi(\tau_0)$.

Relaxed optimization: $\max_{\mathcal{D}} \mathcal{Q}_t \overset{\text{exact symmetrization}}{=} \max_{\mathcal{K}} F_t = \Phi(t)$.

Lower approximation: $G_N \in \mathscr{C},\ N \to \infty \Longrightarrow \Lambda(\mathscr{C}) \ge \Phi(\tau_0)$,
then blow up each fixed template and let the host order $n \to \infty$.

Specialization: Φ semialgebraic and strictly increasing, $\tau_0 = 1/(2e^2) \Longrightarrow \Phi(\tau_0)$ transcendental.

Here $k = 1000$ is fixed before every displayed limit. The index ℓ is only a maximizing-sequence index for the weighted supremum, N is only the Ferrers discretization order, and n is only the order of an ordinary host hypergraph or blow-up. These limits are never interchanged inside a single argument: the lower bound first obtains weighted templates as $N \to \infty$ and then uses a diagonal choice of fixed-template blow-ups as $n \to \infty$.

Object and quantifier ledger

Object or index	Logical status	Allowed conclusion and prohibited inference
k, q, L, J	Fixed integers	Fixed before $\ell \to \infty$, $N \to \infty$ and $n \to \infty$; neither e nor τ_0 enters the forbidden family.
G or $G^{(\ell)}$	Actual finite k-graph in $\mathscr{C}$ with a legal marker map	Graph membership and local rules (R1)–(R6) are available.
$\widehat{G}$	Actual saturated finite supergraph in $\mathscr{C}$	Claim 6.3 gives $\lambda(G) \le \lambda(\widehat{G})$ and the bundles needed for the score decomposition.
$(\widehat{G}, \boldsymbol{\omega})$	Actual graph with a minimum-support optimal weighting	On the high-value branch, Lemma 6.4 applies to every positive-weight object, not to zero-weight objects.
G^+	Actual subgraph after deleting zero-weight objects and retaining markers	$G^+ \in \mathscr{C}$ and $\lambda(G^+) = \lambda(\widehat{G})$; every remaining object in a state or payload sum has the asserted type.
Partial state matrix	Extracted only from positive-weight row and column objects of G^+	Rule (R4) excludes directed 4-cycles. Missing and repaired cells are numerical bookkeeping, not new graph edges.
$\xi = (y, a, b, E_0, E_1) \in \mathcal{D}$	Numerical relaxed tuple	Lemma 12.1 gives forward score domination. A general point of $\mathcal{D}$, or a new split of (E_0, E_1), need not be graph-realizable.
$(y, s) \in \mathcal{K}$	Symmetric numerical point in a rational simplex	Corollary 10.2 identifies numerical maxima over $\mathcal{D}$ and $\mathcal{K}$; it does not reconstruct a finite template.
G_N, $N \to \infty$	Explicit finite Ferrers templates in $\mathscr{C}$	Independent weighted lower-bound direction; N is not the host order.
Host order $n \to \infty$	Ordinary blow-ups and Turán extremal graphs	The exact transfer theorem converts the unscaled Lagrangian to ordinary density with the factor $k!$.
$\Phi(t)$	Maximum of F_t on $\mathcal{K}$	Numerical value function; semialgebraicity and strict increase are used only after the finite family is fixed.

12 The algebraic envelope

This section has three independent steps. First, the next lemma packages the one-way passage from an actual high-value template to the numerical domain $\mathcal{D}$. Second, Proposition 12.2 combines that upper bound with the explicit finite Ferrers lower templates. Third, the semialgebraic lemmas study the already numerical value function Φ; no graph-realization statement is used in that specialization.

Recall the compact rational simplex $\mathcal{K}$ and the polynomial $F_{k,t}$ from (10.10). For notational convenience write

$$F_t(y,s) = e_k(y) + s e_{k-1}(y_m : m \neq h) + JUsp + \frac{L}{4} W A^2 s^2 + \frac{t}{16} W s^4, \tag{12.1}$$

and

$$\Phi_k(t) := \max_{(y,s)\in\mathcal{K}} F_t(y,s), \qquad \Phi(t) := \Phi_k(t) \qquad (0 \le t \le 1). \tag{12.2}$$

All coefficients in (12.1), including the coefficient of t, are rational. Corollary 10.2 gives

$$\max_{\mathcal{D}} \mathcal{Q}_t = \Phi(t) \qquad (0 \le t \le 1).$$

Lemma 12.1 (Template-to-relaxation domination)**.** *Let $\widehat{G} \in \mathscr{C}$ carry a saturated legal marker map, and let $\boldsymbol{\omega}$ be an optimal weighting of $\widehat{G}$ whose positive support has minimum cardinality. If*

$$\lambda(\widehat{G}) > b_q + 10q^{-k},$$

then there is a tuple $\xi = (y, a, b, E_0, E_1) \in \mathcal{D}$ such that

$$\lambda(\widehat{G}) \le \mathcal{Q}_{\tau_0}(\xi) \le \max_{\mathcal{D}} \mathcal{Q}_{\tau_0} = \Phi(\tau_0).$$

The first inequality is only a numerical score domination; the repaired matrix defining ξ need not be realized by $\widehat{G}$ or by any finite template.

Proof. Lemma 6.4 gives the pre-repair tube, $y_h > 0$, and positivity of every marker coordinate. It also gives the required row/column types for every positive-weight object, stated pair and payload. Delete every zero-weight object while retaining all marked vertices, and call the resulting actual subgraph G^+. By Lemma 3.1, $G^+ \in \mathscr{C}$. Restricting $\boldsymbol{\omega}$ preserves its score and hence gives $\lambda(G^+) \ge \lambda(\widehat{G})$; conversely, every weighting of $G^+ \subseteq \widehat{G}$ extends by zero. Therefore

$$\lambda(G^+) = \lambda(\widehat{G}).$$

Thus every object entering the remaining state or payload sums has positive weight and has exactly one of the two asserted types.

Let a and b be the total weights of the row and column objects of G^+. Because every remaining object has exactly one of these types, $a + b$ is the total object weight. Form the partial state matrix on precisely this positive object support. Rule (R4) excludes directed 4-cycles. The rational estimate (7.5) verifies the hypothesis of Lemma 7.1, so that lemma replaces the partial matrix by a complete Ferrers matrix without decreasing the combined state and payload score. Let E_0, E_1 be the total product weights of its state-0 and state-1 cells. Then

$$E_0, E_1 \ge 0, \qquad E_0 + E_1 = ab.$$

The marker weights and the totals a, b are unchanged. Since the object weights together with the marker weights still sum to one, the tuple $\xi = (y, a, b, E_0, E_1)$ belongs to $\mathcal{D}$.

Saturation makes the base, tag, and declared-state terms in (6.5) exact before repair. The repaired state score is $E_0 Z_0 + E_1 Z_1$, and Corollary 5.2 bounds the completed payload score by $\tau_0 W a^2 b^2$. Hence

$$\lambda(\widehat{G}) = \lambda(G^+) \le e_k(x) + U(aP + bP') + E_0 Z_0 + E_1 Z_1 + \tau_0 W a^2 b^2 = \mathcal{Q}_{\tau_0}(\xi).$$

The remaining inequalities follow from the definition of the maximum on $\mathcal{D}$ and Corollary 10.2. $\square$

Proposition 12.2 (Exact weighted extremal identity)**.** *For the fixed integer $k = k_\star$,*

$$\Lambda(\mathscr{C}) = \Phi(\tau_0), \qquad \tau_0 = \frac{1}{2e^2}. \tag{12.3}$$

Proof. The finite Ferrers templates of Section 8 give $\Lambda(\mathscr{C}) \ge \Phi(\tau_0)$ by (8.4). For the reverse inequality, choose $G^{(\ell)} \in \mathscr{C}$ such that

$$\lambda(G^{(\ell)}) \longrightarrow \Lambda(\mathscr{C}),$$

and fix one legal marker map for each graph. Claim 6.3 gives saturated extensions $\widehat{G}^{(\ell)} \in \mathscr{C}$ with

$$\lambda(G^{(\ell)}) \le \lambda(\widehat{G}^{(\ell)}) \le \Lambda(\mathscr{C}),$$

so the middle terms have the same limit. Choose on each saturated graph an optimal weighting of minimum positive support.

The tagged test (6.9) gives

$$\Lambda(\mathscr{C}) \ge b_q + \frac{101}{4} q^{-k} > b_q + 10q^{-k}.$$

Hence, for all sufficiently large ℓ, the hypotheses of Lemma 12.1 hold and

$$\lambda(\widehat{G}^{(\ell)}) \le \Phi(\tau_0).$$

Letting $\ell \to \infty$ gives $\Lambda(\mathscr{C}) \le \Phi(\tau_0)$ and proves the identity. □

We next isolate the semialgebraic fact needed for specialization. In the numbering of the 1998 English edition of Bochnak–Coste–Roy, Nash stratification and the semialgebraic Sard theorem [4, Proposition 9.1.8 and Theorem 9.6.2], together with the characterization of smooth semialgebraic functions as Nash functions [4, Proposition 8.1.8], give the standard piecewise-Nash description of a one-variable semialgebraic function away from finitely many critical values. Thus, conceptually, its graph is piecewise contained in algebraic curves. For the specialization argument we additionally need the defining curve over $\mathbb{Q}$ and every exceptional endpoint algebraic over $\mathbb{Q}$. We therefore include the elementary factor-and-boundary proof below to track the field of definition explicitly.

We use the Tarski–Seidenberg theorem, stability under closure and boundary, stability under semialgebraic images and inverse images, and the finiteness and semialgebraicity of connected components in the precise forms [4, Theorem 2.2.1, Propositions 2.2.2 and 2.2.7, and Theorem 2.4.5]. All defining coefficients below lie in $\mathbb{Q}$ (or, in one auxiliary level-set argument, in a finite algebraic extension of $\mathbb{Q}$).

Lemma 12.3 (Rational algebraic arcs of a semialgebraic function)**.** *Let $f : I \to \mathbb{R}$ be a function whose graph is semialgebraic over $\mathbb{Q}$, where I is an interval. There is a finite set $B \subset I$ of algebraic numbers such that, on every component J of $I \setminus B$, the graph of $f|_J$ is contained in an irreducible algebraic curve*

$$Q(T, Z) = 0, \qquad Q \in \mathbb{Q}[T, Z].$$

If f is nonconstant on J, then Q can be chosen to depend on both T and Z.

Proof. Let

$$\Gamma = \{(t, f(t)) : t \in I\}.$$

If $I = \varnothing$, take $B = \varnothing$. If I has dimension zero and is nonempty, then $I = \{t_0\}$. The projection I of Γ is semialgebraic over $\mathbb{Q}$ by the Tarski–Seidenberg theorem [4, Theorem 2.2.1]. Choose a quantifier-free formula over $\mathbb{Q}$ defining this singleton. Some nonzero polynomial occurring in that formula must vanish at t_0: otherwise all occurring signs would be locally constant, so the formula would hold either on a neighbourhood of t_0 or nowhere there. Thus t_0 is algebraic, and the assertion follows with $B = \{t_0\}$. Assume henceforth that I is nondegenerate. By quantifier elimination, membership in Γ is determined by the signs of finitely many nonzero polynomials in $\mathbb{Q}[T, Z]$. Partition $\mathbb{R}^2$ into the corresponding sign cells. Every nonempty sign cell on which the defining formula for Γ is true is contained in the graph Γ. Such a cell cannot have all signs strict, since then it would be open in $\mathbb{R}^2$, whereas a graph contains no nonempty open set. Hence at least one of the defining polynomials vanishes on each such cell. Choosing one for every realized cell and taking their product gives a nonzero polynomial

$$P(T, Z) \in \mathbb{Q}[T, Z]$$

that vanishes on all of Γ.

Factor P in $\mathbb{Q}[T, Z]$ as

$$P = c \prod_{i=1}^{r} Q_i^{m_i},$$

where the Q_i are pairwise nonassociate irreducible polynomials. For each i put

$$S_i = \{t \in I : Q_i(t, f(t)) = 0\}.$$

The graph map $t \mapsto (t, f(t))$ is semialgebraic, so every S_i is semialgebraic over $\mathbb{Q}$ by stability under semialgebraic inverse images [4, Proposition 2.2.7], and $I = \bigcup_i S_i$. By [4, Proposition 2.2.2 and Theorem 2.4.5], a semialgebraic subset of the real line has finite semialgebraic boundary. We also record why the field of definition controls its points. Take a quantifier-free formula over $\mathbb{Q}$ for S_i and let α be a boundary point. If no nonzero polynomial occurring in the formula vanished at α, all of their signs would be locally constant, so membership in S_i would be locally constant, contradicting that α is a boundary point. Thus α is a zero of a nonzero polynomial in $\mathbb{Q}[T]$ and is algebraic. Put all these boundary points in B.

There is one harmless refinement needed for factors depending only on Z. For every i with $Q_i \in \mathbb{Q}[Z]$ and every real root ζ of Q_i, add to B the boundary points in I of the level set $\{t : f(t) = \zeta\}$. These sets are semialgebraic over the number field $\mathbb{Q}(\zeta)$. The same sign-stability argument shows that each boundary point is algebraic over $\mathbb{Q}(\zeta)$, and hence algebraic over $\mathbb{Q}$. Thus B remains finite and algebraic.

Let J be a component of $I \setminus B$. For every i, the set $S_i \cap J$ has empty relative boundary in the connected interval J; hence it is either empty or all of J. Since the S_i cover I, at least one index i satisfies $S_i \cap J = J$. The graph of $f|_J$ is then contained in the irreducible curve $Q_i(T, Z) = 0$.

A factor depending only on T cannot vanish on a nonempty interval J. If every factor that contains the graph over J depended only on Z, the level-set refinement in the definition of B would force f to be constant on J. Therefore, whenever f is nonconstant on J, one may choose an applicable factor Q_i that depends on both variables. $\square$

Lemma 12.4. *The function Φ is semialgebraic over $\mathbb{Q}$ and is strictly increasing on* $[0, 1]$.

Proof. Its graph is defined in the ordered field of reals by

$$z = \Phi(t) \quad \iff \quad \exists (y, s) \in \mathcal{K} \left(z = F_t(y, s) \ \wedge \ \forall (y', s') \in \mathcal{K}\ F_t(y', s') \le z \right).$$

Tarski–Seidenberg quantifier elimination [4, Theorem 2.2.1] therefore makes the graph semialgebraic over $\mathbb{Q}$.

For every t, the tagged test weighting from (6.9) gives

$$\Phi(t) > b_q.$$

Consequently a maximizing point has $s > 0$, since for $s = 0$ all terms after the base vanish and $e_k(y) \le b_q$. By Corollary 10.2, this point lifts (with $a = b = s/2$ and a suitable split $E_0 + E_1 = ab$) to a maximizer of $\mathcal{Q}_t$ on $\mathcal{D}$. Lemma 9.1 therefore gives $y_m > 0$ for every $m \neq h$, in particular for every $m \in D$; hence $W > 0$. If $0 \le t_1 < t_2 \le 1$ and (y_1, s_1) maximizes F_{t_1}, then

$$\Phi(t_2) \ge F_{t_2}(y_1, s_1) = \Phi(t_1) + \frac{t_2 - t_1}{16} W(y_1) s_1^4 > \Phi(t_1).$$

Thus Φ is strictly increasing. $\square$

13 Transcendence

Theorem 13.1. *The two numbers $\Phi(1/(2e^2))$ and $k!\Phi(1/(2e^2))$ are transcendental over $\mathbb{Q}$.*

Proof. We use the Lindemann–Weierstrass theorem in the following precise form [3, Chapter 1, §3]: if $0 \neq \gamma \in \overline{\mathbb{Q}}$, then e^{γ} is transcendental. Taking $\gamma = -2$ shows that e^{-2} is transcendental. Therefore

$$\tau_0 = \frac{1}{2}e^{-2}$$

is transcendental as well, since otherwise $e^{-2} = 2\tau_0$ would be algebraic.

Apply Lemma 12.3 to Φ. Every point of the finite exceptional set B is algebraic, so $\tau_0 \notin B$. Hence on an open interval about τ_0 there is an irreducible polynomial $Q(T, Z) \in \mathbb{Q}[T, Z]$, depending on both variables, such that

$$Q(t, \Phi(t)) = 0. \tag{13.1}$$

The branch is nonconstant by Lemma 12.4.

Suppose that $\beta := \Phi(\tau_0)$ were algebraic, and write

$$Q(T, Z) = \sum_{j=0}^{d} a_j(Z)T^j, \qquad a_j \in \mathbb{Q}[Z].$$

Then $Q(T, \beta)$ is not the zero polynomial. Indeed, let $m_\beta \in \mathbb{Q}[Z]$ be the minimal polynomial of β. The kernel of the evaluation homomorphism $\mathbb{Q}[Z] \to \mathbb{Q}(\beta)$ is the principal ideal (m_β). Thus, if $Q(T, \beta)$ vanished identically, m_β would divide every a_j, and hence would divide Q in $\mathbb{Q}[T, Z]$. Since Q is irreducible, it would then be associated to $m_\beta(Z)$ and could not depend on T, contrary to the choice of the curve. Now

$$Q(\tau_0, \beta) = 0$$

exhibits τ_0 as a root of the nonzero polynomial $Q(T, \beta) \in \mathbb{Q}(\beta)[T]$. Hence τ_0 is algebraic over the finite algebraic extension $\mathbb{Q}(\beta)$, and therefore algebraic over $\mathbb{Q}$. This contradiction proves that $\Phi(\tau_0)$ is transcendental. Multiplication by the nonzero integer $k!$ preserves transcendence. $\square$

Combining Theorem 4.4, (12.3), and Theorem 13.1 gives

$$\pi(\mathcal{F}) = k!\Phi(1/(2e^2)),$$

and this is a transcendental classical Turán density of one fixed finite family of finite simple k-graphs.

14 Completion of the proof and audits

We collect the logical chain, including the quantifiers that are easy to lose in this problem.

Proof of Theorem 1.1. Take the fixed integer $k = k_\star$ from (11.6), form the literal profile class $\mathscr{C}$, its finite obstruction family $\mathcal{A}$ from Proposition 3.3, and the finite core-completion family $\mathcal{F}$ from Section 4. The weighted extremal analysis gives

$$\Lambda(\mathscr{C}) = \Phi(\tau_0), \qquad \tau_0 = \frac{1}{2e^2}.$$

Theorem 4.4 therefore gives

$$\pi(\mathcal{F}) = k!\Phi(\tau_0).$$

The right side is transcendental by Theorem 13.1. This proves the theorem. $\square$

Finiteness audit. Every object in $\mathcal{F}$ is an ordinary finite simple k-graph. The obstruction family $\mathcal{A}$ is a subset of the finite set of k-graphs on at most

$$9k\sum_{d=0}^{q}(9kq)^d$$

vertices. For $A \in \mathcal{A}$, every A-core completion has at most

$$|V(A)| + (k-2)\binom{|V(A)|}{2}$$

vertices, so $\mathcal{F}$ is finite. Marker labels, states and tags are used only to define $\mathcal{A}$; none remains in a forbidden graph.

Parameter audit. The family is fixed after the integer $k = 1000$ is certified by (11.1)–(11.5). These are finitely many rational inequalities; Appendix A gives their exact integer cross-multiplications and the companion verification script. Neither $\mathcal{A}$ nor $\mathcal{F}$ depends on the host order n, a numerical precision, the Ferrers discretization order, the number e, or the value τ_0.

Lower-bound audit. The finite Ferrers templates $G_N \in \mathscr{C}$ give weighted values tending to $\Phi(\tau_0)$. Rounded blow-ups of each fixed template are $\mathcal{F}$-free by Lemma 4.1. A diagonal choice of template and blow-up order yields

$$\liminf_{n\to\infty} \frac{\mathrm{ex}(n,\mathcal{F})}{\binom{n}{k}} \geq k!\Phi(\tau_0).$$

Upper-bound audit. Let H be any $\mathcal{F}$-free k-graph, with no structural assumption. A minimum-support optimal weighting makes its support pair-covered. A copy of an obstruction A on that support would extend inside H to an A-core completion, contradicting $\mathcal{F}$-freeness. Thus $\lambda(H) \leq \Phi(\tau_0)$. At the uniform weighting,

$$\frac{|E(H)|}{n^k} \leq \lambda(H),$$

and consequently

$$\limsup_{n\to\infty} \frac{\mathrm{ex}(n,\mathcal{F})}{\binom{n}{k}} \leq k!\Phi(\tau_0).$$

This applies to every $\mathcal{F}$-free host and along the full limit.

Object-level audit. The saturation and zero-weight deletion steps remain inside the actual hereditary class $\mathscr{C}$. The subsequent matrix repair is used only to produce a score-dominating relaxed tuple in $\mathcal{D}$; neither the repair nor the redistribution of (E_0, E_1) is asserted to be realized in the same finite graph. Equality of the maxima over $\mathcal{D}$ and $\mathcal{K}$ is a numerical variational identity. The reverse, lower-bound direction is supplied separately by the explicit finite Ferrers templates $G_N \in \mathscr{C}$.

Normalization and transcendence audit. Since $\binom{n}{k} = n^k/k! + O_k(n^{k-1})$, the factor $k!$ in the density is exact. The Ferrers calculation gives exactly, not approximately, $\tau_0 = 1/(2e^2)$. The Lindemann–Weierstrass theorem [3, Chapter 1, §3] proves that τ_0 is transcendental. Semialgebraic cell decomposition supplies an algebraic curve over $\mathbb{Q}$ through the graph of the strictly increasing function Φ near τ_0; specialization then proves that $\Phi(\tau_0)$, and hence $k!\Phi(\tau_0)$, is transcendental.

We have therefore established the full asymptotic statement

$$\mathrm{ex}(n,\mathcal{F}) = \big(k!\Phi(1/(2e^2)) + o(1)\big)\binom{n}{k}, \qquad k!\Phi(1/(2e^2)) \notin \overline{\mathbb{Q}}.$$

A Exact arithmetic certificate for $k = 1000$

All numerical assertions used to freeze the construction are rational. For $k = 1000$, $q = 10^9$, $L = 4$ and $J = 100$, the five inequalities (11.1)–(11.5) are respectively equivalent to the following exact integer comparisons:

$$\begin{aligned} 120k!q^2 &< k(k-1)k^k, \\ 30 \cdot 10^{12} k^2 q^{k+4} &< k^k \binom{q}{k}, \\ 4 \cdot 10^8 k^4 &< \binom{q-2}{k-2}, \\ 16J(k-2)! &< Lq^{k-2}, \\ 49(10^9+1)^{1000} &< 240(10^9)^{1000}. \end{aligned}$$

The last line is obtained from $(L/2+1/24)(1+1/(10^6 k))^k < 10$ without any floating-point evaluation. The reserved-label check is the integer comparison $2k - 6 + 2L + 2J = 2202 < 10^9$.

The submission archive contains the companion files

`k1000_exact_certificate.py`, `k1000_exact_certificate_output.txt`, `README.md`, `MANIFEST.sha256`.

The recorded output is reproduced from a clean run with Python 3.13.5 by the command

```
python3 k1000_exact_certificate.py >
k1000_exact_certificate_output.txt.
```

The script has no third-party dependency and uses only Python's arbitrary-precision integers and `fractions.Fraction`. It checks the five displayed comparisons, the product estimates used in Lemma 11.1, the repair margin $81/25 > 297/200$, and the tagged-test identity $J/4 + L/16 = 101/4$. The output records all fifteen checks as `PASS`; floating-point logarithms in its final diagnostic block are not used as proofs. The exact reproduction command, the tested TeX and Python versions, and SHA-256 hashes of every archived source and certificate file are recorded in `README.md` and `MANIFEST.sha256`.

References

[1] J. Babu, A. S. Jacob, R. Krithika and D. Rajendraprasad, *A note on arc-disjoint cycles in bipartite tournaments*, arXiv:2002.06912v1, 17 February 2020.

[2] R. Baber and J. Talbot, *New Turán densities for 3-graphs*, Electron. J. Combin. **19** (2012), no. 2, Paper P22, 21 pp.

[3] A. Baker, *Transcendental Number Theory*, Cambridge Mathematical Library, Cambridge University Press, Cambridge, 1990.

[4] J. Bochnak, M. Coste and M.-F. Roy, *Real Algebraic Geometry*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3rd series, vol. 36, Springer-Verlag, Berlin, first English edition, 1998, doi:10.1007/978-3-662-03718-8.

[5] C. Grosu, *On the algebraic and topological structure of the set of Turán densities*, J. Combin. Theory Ser. B **118** (2016), 137–185.

[6] J. Hou, H. Li, X. Liu, D. Mubayi and Y. Zhang, *Hypergraphs with infinitely many extremal constructions*, Discrete Anal. (2023), Paper No. 18, 34 pp.

[7] G. Katona, T. Nemetz and M. Simonovits, *On a problem of Turán in the theory of graphs*, Mat. Lapok **15** (1964), 228–238.

[8] X. Liu and O. Pikhurko, *Hypergraph Turán densities can have arbitrarily large algebraic degree*, J. Combin. Theory Ser. B **161** (2023), 407–416.

[9] X. Liu and O. Pikhurko, *Intervals of hypergraph Turán densities*, arXiv:2605.25914v1, 25 May 2026.

[10] O. Pikhurko, *On possible Turán densities*, Israel J. Math. **201** (2014), no. 1, 415–454.

[11] R. Yuster, *On the maximum density of a matrix and a transcendental Turán-type density*, arXiv:2601.14904v1, 21 January 2026.